Chapter 12

Vectors

Contents:

A Vectors and scalars
B Geometric operations with vectors
C Vectors in the plane
D The magnitude of a vector
E Operations with plane vectors
F Vectors in space
G Operations with vectors in space
H Vector algebra
I The vector between two points
J Parallelism
K The scalar product of two vectors
L The angle between two vectors
M Proof using vector geometry
N The vector product of two vectors



OPENING PROBLEM

An aeroplane in calm conditions is flying at 800 km h^{-1} due east. A cold wind suddenly blows from the south-west at 35 km h^{-1}, pushing the aeroplane slightly off course.

Things to think about:

a How can we illustrate the plane's movement and the wind using a scale diagram?

b What operation do we need to perform to find the effect of the wind on the aeroplane?

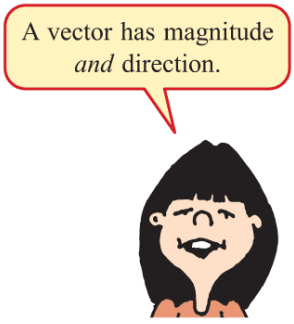
c Can you use a scale diagram to determine the resulting speed and direction of the aeroplane?

A VECTORS AND SCALARS

In the **Opening Problem**, the effect of the wind on the aeroplane is determined by both its speed *and* its direction. The effect would be different if the wind was blowing against the aeroplane rather than from behind it.

Quantities which have only magnitude are called **scalars**.

Quantities which have both magnitude and direction are called **vectors**.



The *speed* of the plane is a scalar. It describes how fast the plane is moving.

The *velocity* of the plane is a vector. It includes both its speed and also its direction.

Other examples of vector quantities are:

- acceleration
- displacement
- force
- momentum



For example, farmer Giles needs to remove a fence post. He starts by pushing on the post sideways to loosen the ground. Giles has a choice of how hard to push the post and in which direction. The force he applies is therefore a vector.

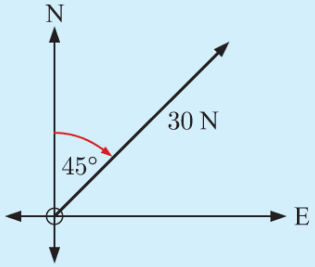
DIRECTED LINE SEGMENT REPRESENTATION

We can represent a vector quantity using a **directed line segment** or **arrow**.

The **length of the arrow** represents the size or magnitude of the quantity, and the **arrowhead** shows its direction.

Example 1 **Self Tutor**

Draw a scale diagram to represent a force of 30 newtons in a north-easterly direction.



Scale:
1 cm ≡ 10 N

EXERCISE 12A.1

1 Using a scale of 1 cm represents 10 units, sketch a vector to represent:

a 30 newtons in a south-easterly direction

b 25 m s^{-1} in a northerly direction

c an excavator digging a tunnel at a rate of 30 cm min^{-1} at an angle of 30° to the ground

d an aeroplane taking off at an angle of 10° to the runway with a speed of 50 m s^{-1}.

2 If $\xrightarrow{25\text{ mm}}$ represents a velocity of 50 m s^{-1} due east, draw a directed line segment representing a velocity of:

a 100 m s^{-1} due west **b** 75 m s^{-1} north-east.

3 Draw a scale diagram to represent:

a a force of 30 newtons in a north-westerly direction

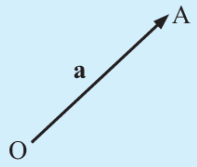
b a velocity of 36 m s^{-1} vertically downwards

c a displacement of 4 units at an angle of 15° to the positive x-axis

d the velocity of an aeroplane taking off at an angle of 8° to the runway at a speed of 150 km h^{-1}.

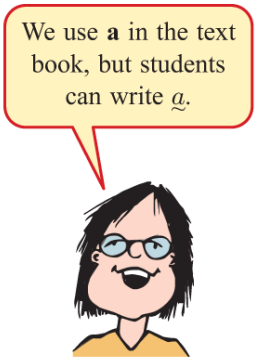
VECTOR NOTATION

- The vector from the origin O to point A is called the **position vector** of point A. It is written as $\overrightarrow{OA}$, **a** or $\underset{\sim}{a}$. The **magnitude** or **length** of vector $\overrightarrow{OA}$ is written as $|\overrightarrow{OA}|$ or OA or $|\mathbf{a}|$ or $|\underset{\sim}{a}|$.



- $\overrightarrow{AB}$ is the vector which **originates** at A and **terminates** at B, $\overrightarrow{AB}$ is called the **displacement vector** of B relative to A.



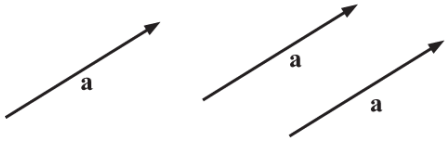


GEOMETRIC VECTOR EQUALITY

Two vectors are **equal** if they have the same magnitude *and* direction.

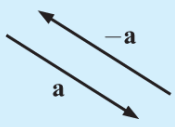
Equal vectors are **parallel** and **in the same direction**, and are **equal in length**. The arrows that represent them are translations of one another.

We can draw a vector with given magnitude and direction from *any* point, so we consider vectors to be **free**.



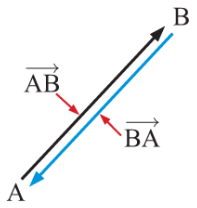
GEOMETRIC NEGATIVE VECTORS

The **negative** of a vector **a** is the vector parallel to **a** and with the same length, but in the opposite direction. It is denoted $-\mathbf{a}$.

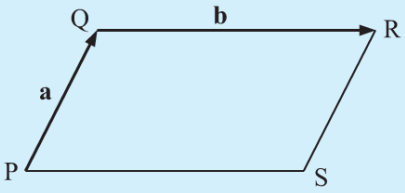


For example, $\overrightarrow{AB}$ and $\overrightarrow{BA}$ have the same length, but they have opposite directions.

$\overrightarrow{BA}$ is the **negative** of $\overrightarrow{AB}$, and we write $\overrightarrow{BA} = -\overrightarrow{AB}$.



Example 2 **Self Tutor**



PQRS is a parallelogram in which $\overrightarrow{PQ} = \mathbf{a}$ and $\overrightarrow{QR} = \mathbf{b}$.

Find vector expressions for:

a $\overrightarrow{QP}$ **b** $\overrightarrow{RQ}$ **c** $\overrightarrow{SR}$ **d** $\overrightarrow{SP}$

a $\overrightarrow{QP} = -\mathbf{a}$ {the negative vector of $\overrightarrow{PQ}$}

b $\overrightarrow{RQ} = -\mathbf{b}$ {the negative vector of $\overrightarrow{QR}$}

c $\overrightarrow{SR} = \mathbf{a}$ {parallel to and the same length as $\overrightarrow{PQ}$}

d $\overrightarrow{SP} = -\mathbf{b}$ {parallel to and the same length as $\overrightarrow{QR}$, but opposite in direction}

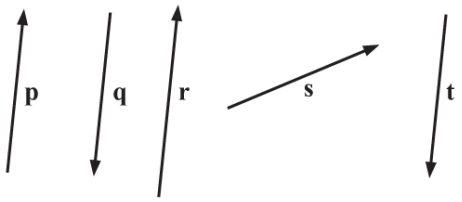
EXERCISE 12A.2

1 State the vectors which are:

a equal in magnitude **b** parallel

c in the same direction **d** equal

e negatives of one another.



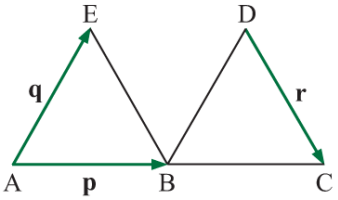
2 The figure shown consists of two equilateral triangles. A, B, and C are collinear.

$\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{AE} = \mathbf{q}$, and $\overrightarrow{DC} = \mathbf{r}$.

Which of the following statements are true?

a $\overrightarrow{EB} = \mathbf{r}$ **b** $|\mathbf{p}| = |\mathbf{q}|$ **c** $\overrightarrow{BC} = \mathbf{r}$

d $\overrightarrow{DB} = \mathbf{q}$ **e** $\overrightarrow{ED} = \mathbf{p}$ **f** $\mathbf{p} = \mathbf{q}$



3 ABCDEF is a regular hexagon.

a Write down the vector which:

i originates at B and terminates at C

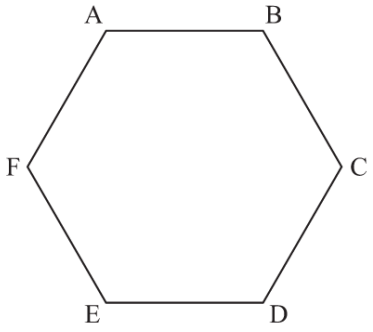
ii is equal to $\overrightarrow{AB}$.

b Write down *all* vectors which:

i are the negative of $\overrightarrow{EF}$

ii have the same length as $\overrightarrow{ED}$.

c Write down a vector which is parallel to $\overrightarrow{AB}$ and twice its length.



DISCUSSION

- Does a vector of length 0 have a direction?
- Is it possible to represent such a vector geometrically?

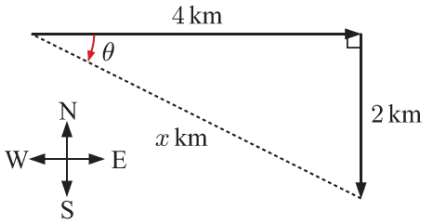
B GEOMETRIC OPERATIONS WITH VECTORS

In previous years we have often used trigonometry and Pythagoras' theorem to solve problems involving distances and directions.

A typical problem could be:

A runner runs east for 4 km and then south for 2 km.

How far is she from her starting point and in what direction?



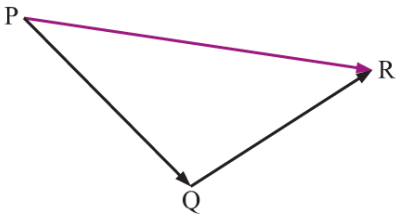
Problems like these can be well represented by vectors. In this case, the vectors are displacements. There is a displacement of 4 km to the east followed by a displacement of 2 km to the south. We want to know the *sum* of these displacements, including distance *and* direction.

GEOMETRIC VECTOR ADDITION

Suppose we have three towns P, Q, and R.

A trip from P to Q followed by a trip from Q to R has the same origin and destination as a trip from P to R.

This can be expressed in vector form as the sum $\overrightarrow{PQ} + \overrightarrow{QR} = \overrightarrow{PR}$.



From examples like this, we can define vector addition geometrically as follows:

To construct $\mathbf{a} + \mathbf{b}$:

Step 1: Draw $\mathbf{a}$.

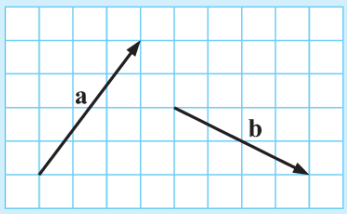
Step 2: At the arrowhead end of $\mathbf{a}$, draw $\mathbf{b}$.

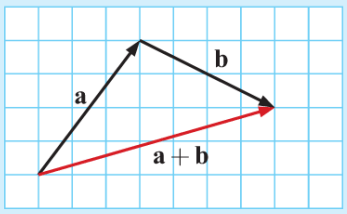
Step 3: Join the beginning of $\mathbf{a}$ to the arrowhead end of $\mathbf{b}$. This is vector $\mathbf{a} + \mathbf{b}$.



Example 3 Self Tutor

Given vectors $\mathbf{a}$ and $\mathbf{b}$ shown, construct $\mathbf{a} + \mathbf{b}$.





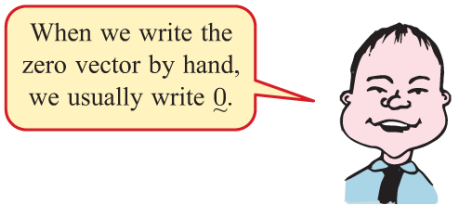
THE ZERO VECTOR

Having defined vector addition, we are now able to state that:

The **zero vector 0** is a vector of length 0.

For any vector **a**: $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$

$\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$.



Example 4 **Self Tutor**

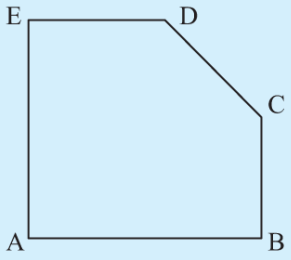
Find a single vector which is equal to:

a $\overrightarrow{BC} + \overrightarrow{CA}$

b $\overrightarrow{BA} + \overrightarrow{AE} + \overrightarrow{EC}$

c $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}$

d $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE}$



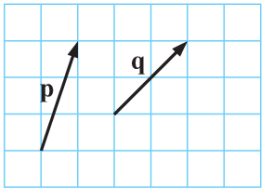
a $\overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{BA}$ **b** $\overrightarrow{BA} + \overrightarrow{AE} + \overrightarrow{EC} = \overrightarrow{BC}$

c $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AA} = \mathbf{0}$ **d** $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{AE}$

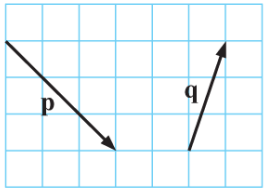
EXERCISE 12B.1

1 Use the given vectors **p** and **q** to construct $\mathbf{p} + \mathbf{q}$:

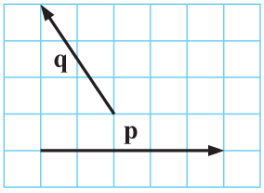
a



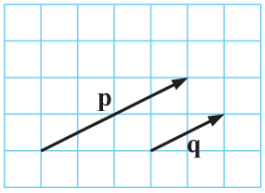
b



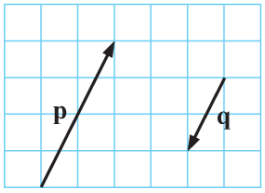
c



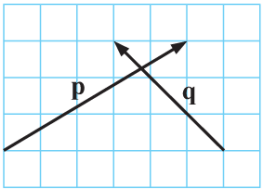
d



e



f

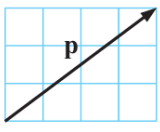


2 Find a single vector which is equal to:

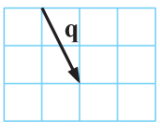
a $\overrightarrow{AB} + \overrightarrow{BC}$ **b** $\overrightarrow{BC} + \overrightarrow{CD}$ **c** $\overrightarrow{AB} + \overrightarrow{BA}$

d $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$ **e** $\overrightarrow{AC} + \overrightarrow{CB} + \overrightarrow{BD}$ **f** $\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB}$

3 **a** Given



and

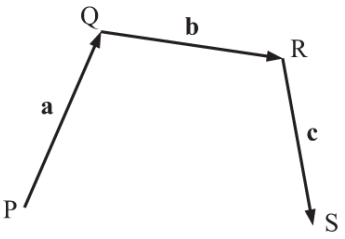


use vector diagrams to find:

i $\mathbf{p} + \mathbf{q}$ **ii** $\mathbf{q} + \mathbf{p}$.

b For any two vectors **p** and **q**, is $\mathbf{p} + \mathbf{q} = \mathbf{q} + \mathbf{p}$? Illustrate your answer.

4 Consider:



One way of finding $\overrightarrow{PS}$ is:

$$\overrightarrow{PS} = \overrightarrow{PR} + \overrightarrow{RS}$$
$$= (\mathbf{a} + \mathbf{b}) + \mathbf{c}.$$

Use the diagram to show that $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$.

5 Answer the **Opening Problem** on page **270**.

6 Susan and Cathy are pushing a heavy trolley containing groceries. Susan pushes the trolley with force 40 N in the direction 017°. Cathy pushes the trolley with force 35 N in the direction 335°.

a Use a scale diagram to estimate the resultant force from the two girls pushing.

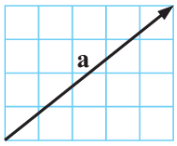
b Use trigonometry to find the resultant force more accurately.

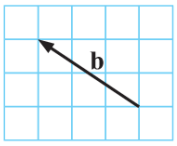


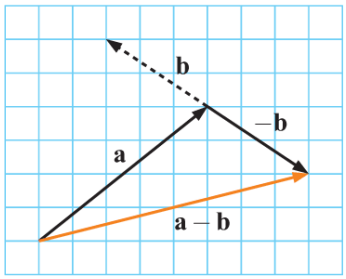
GEOMETRIC VECTOR SUBTRACTION

To subtract one vector from another, we simply **add its negative**. $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$

For example,

given 

and 

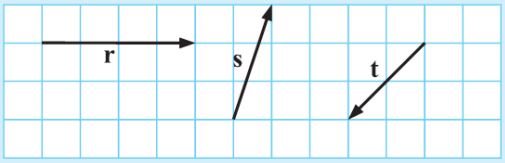
then 

Example 5 **Self Tutor**

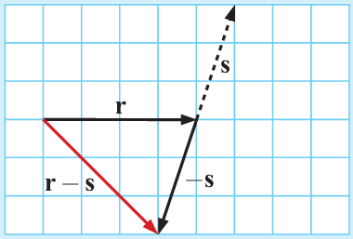
For the vectors illustrated, show how to construct:

a $\mathbf{r} - \mathbf{s}$

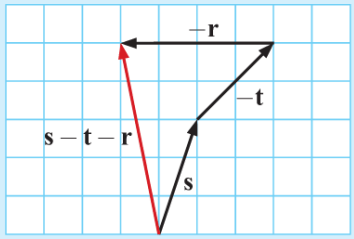
b $\mathbf{s} - \mathbf{t} - \mathbf{r}$



a



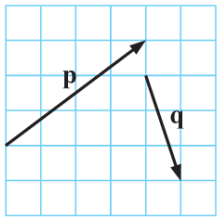
b



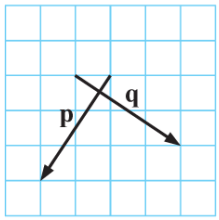
EXERCISE 12B.2

1 For the following vectors $\mathbf{p}$ and $\mathbf{q}$, show how to construct $\mathbf{p} - \mathbf{q}$:

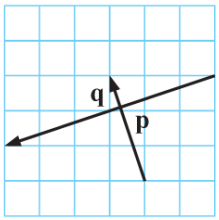
a



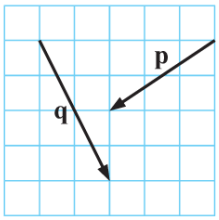
b



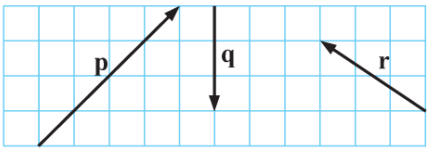
c



d



2 For the vectors illustrated, show how to construct:



a $\mathbf{p} + \mathbf{q} - \mathbf{r}$ **b** $\mathbf{p} - \mathbf{q} - \mathbf{r}$ **c** $\mathbf{r} - \mathbf{q} - \mathbf{p}$

Example 6 Self Tutor

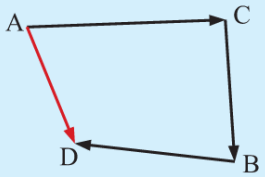
For points A, B, C, and D, simplify the following vector expressions:

a $\overrightarrow{AB} - \overrightarrow{CB}$ **b** $\overrightarrow{AC} - \overrightarrow{BC} - \overrightarrow{DB}$

a
$$\begin{aligned} &\overrightarrow{AB} - \overrightarrow{CB} \\ &= \overrightarrow{AB} + \overrightarrow{BC} \quad \{\text{as } \overrightarrow{BC} = -\overrightarrow{CB}\} \\ &= \overrightarrow{AC} \end{aligned}$$



b
$$\begin{aligned} &\overrightarrow{AC} - \overrightarrow{BC} - \overrightarrow{DB} \\ &= \overrightarrow{AC} + \overrightarrow{CB} + \overrightarrow{BD} \\ &= \overrightarrow{AD} \end{aligned}$$



3 For points A, B, C, and D, simplify the following vector expressions:

a $\overrightarrow{AC} + \overrightarrow{CB}$ **b** $\overrightarrow{AD} - \overrightarrow{BD}$ **c** $\overrightarrow{AC} + \overrightarrow{CA}$

d $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$ **e** $\overrightarrow{BA} - \overrightarrow{CA} + \overrightarrow{CB}$ **f** $\overrightarrow{AB} - \overrightarrow{CB} - \overrightarrow{DC}$

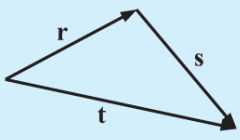
VECTOR EQUATIONS

Whenever we have vectors which form a closed polygon, we can write a **vector equation**. The vector equation can usually be written in several ways, but they are all equivalent.

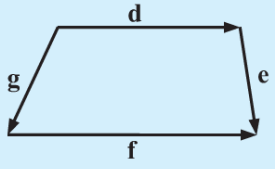
Example 7 Self Tutor

Construct vector equations for:

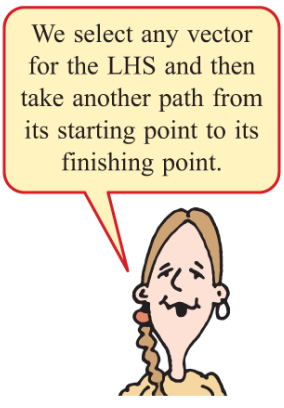
a



b



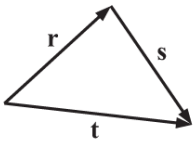
a $\mathbf{t} = \mathbf{r} + \mathbf{s}$ **b** $\mathbf{f} = -\mathbf{g} + \mathbf{d} + \mathbf{e}$



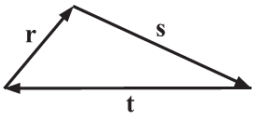
EXERCISE 12B.3

1 Construct vector equations for:

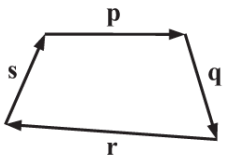
a



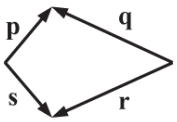
b



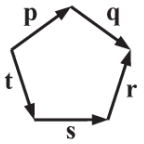
c



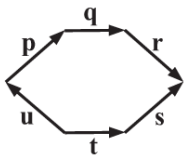
d



e



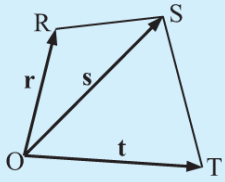
f



Example 8 **Self Tutor**

Find, in terms of **r**, **s**, and **t**:

a $\overrightarrow{RS}$ **b** $\overrightarrow{SR}$ **c** $\overrightarrow{ST}$



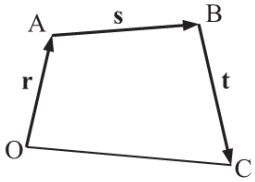
a $\overrightarrow{RS} = \overrightarrow{RO} + \overrightarrow{OS}$
$= -\mathbf{r} + \mathbf{s}$
$= \mathbf{s} - \mathbf{r}$

b $\overrightarrow{SR} = \overrightarrow{SO} + \overrightarrow{OR}$
$= -\mathbf{s} + \mathbf{r}$
$= \mathbf{r} - \mathbf{s}$

c $\overrightarrow{ST} = \overrightarrow{SO} + \overrightarrow{OT}$
$= -\mathbf{s} + \mathbf{t}$
$= \mathbf{t} - \mathbf{s}$

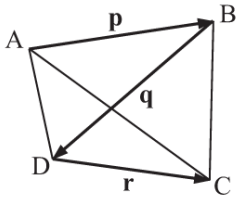
2 **a** Find, in terms of **r**, **s**, and **t**:

i $\overrightarrow{OB}$ **ii** $\overrightarrow{CA}$ **iii** $\overrightarrow{OC}$



b Find, in terms of **p**, **q**, and **r**:

i $\overrightarrow{AD}$ **ii** $\overrightarrow{BC}$ **iii** $\overrightarrow{AC}$



GEOMETRIC SCALAR MULTIPLICATION

A **scalar** is a non-vector quantity. It has magnitude but no direction.

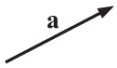
A scalar multiple of a vector is always **parallel** to the vector.



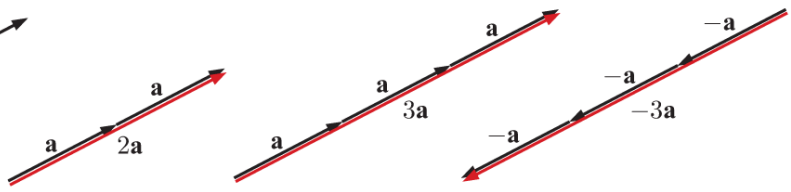
We can multiply vectors by scalars such as 2 and -3, or in fact any $k \in \mathbb{R}$.

If **a** is a vector, we define:

- $2\mathbf{a} = \mathbf{a} + \mathbf{a}$ has the same direction as **a** but is twice the length
- $3\mathbf{a} = \mathbf{a} + \mathbf{a} + \mathbf{a}$ has the same direction as **a** but is three times the length
- $-3\mathbf{a} = 3(-\mathbf{a}) = (-\mathbf{a}) + (-\mathbf{a}) + (-\mathbf{a})$ has the opposite direction to **a** and is three times the length.

If **a** is 

then we have:



If $\mathbf{a}$ is a vector and k is a scalar, then $k\mathbf{a}$ is also a vector and we are performing **scalar multiplication**.

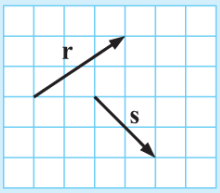
- If $k > 0$, $k\mathbf{a}$ and $\mathbf{a}$ have the same direction.
- If $k < 0$, $k\mathbf{a}$ and $\mathbf{a}$ have opposite directions.
- If $k = 0$, $k\mathbf{a} = \mathbf{0}$, the zero vector.



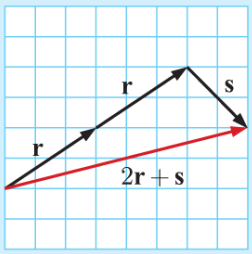
Example 9 Self Tutor

Given the vectors below, construct:

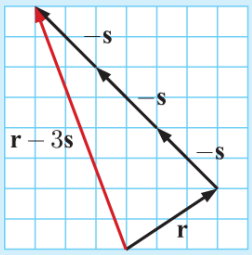
a $2\mathbf{r} + \mathbf{s}$ **b** $\mathbf{r} - 3\mathbf{s}$



a



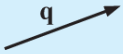
b



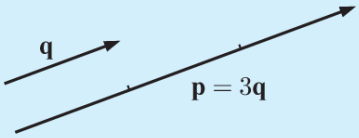
Example 10 Self Tutor

Sketch vectors $\mathbf{p}$ and $\mathbf{q}$ if: **a** $\mathbf{p} = 3\mathbf{q}$ **b** $\mathbf{p} = -\frac{1}{2}\mathbf{q}$

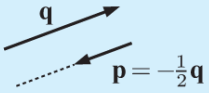
Suppose $\mathbf{q}$ is:



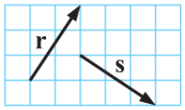
a



b



EXERCISE 12B.4

1 Given vectors 

, construct geometrically:

a $-\mathbf{r}$ **b** $2\mathbf{s}$ **c** $\frac{1}{2}\mathbf{r}$ **d** $-\frac{3}{2}\mathbf{s}$

e $2\mathbf{r} - \mathbf{s}$ **f** $2\mathbf{r} + 3\mathbf{s}$ **g** $\frac{1}{2}\mathbf{r} + 2\mathbf{s}$ **h** $\frac{1}{2}(\mathbf{r} + 3\mathbf{s})$

2 Sketch vectors $\mathbf{p}$ and $\mathbf{q}$ if:

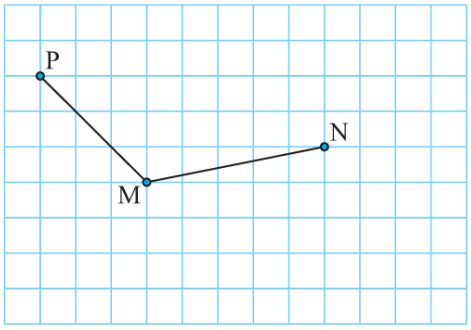
a $\mathbf{p} = \mathbf{q}$ **b** $\mathbf{p} = -\mathbf{q}$ **c** $\mathbf{p} = 2\mathbf{q}$ **d** $\mathbf{p} = \frac{1}{3}\mathbf{q}$ **e** $\mathbf{p} = -3\mathbf{q}$

3 **a** Copy this diagram and on it mark the points:

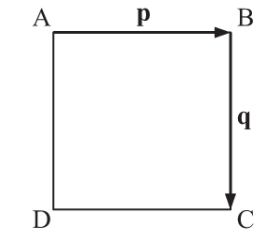
i X such that $\overrightarrow{MX} = \overrightarrow{MN} + \overrightarrow{MP}$

ii Y such that $\overrightarrow{MY} = \overrightarrow{MN} - \overrightarrow{MP}$

iii Z such that $\overrightarrow{PZ} = 2\overrightarrow{PM}$.

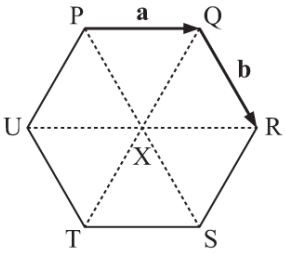


b What type of figure is MNYZ?

4 

ABCD is a square. Its diagonals [AC] and [BD] intersect at M. If $\overrightarrow{AB} = \mathbf{p}$ and $\overrightarrow{BC} = \mathbf{q}$, find in terms of $\mathbf{p}$ and $\mathbf{q}$:

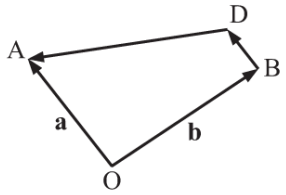
a $\overrightarrow{CD}$ **b** $\overrightarrow{AC}$ **c** $\overrightarrow{AM}$ **d** $\overrightarrow{BM}$

5 

PQRSTU is a regular hexagon.

If $\overrightarrow{PQ} = \mathbf{a}$ and $\overrightarrow{QR} = \mathbf{b}$, find in terms of $\mathbf{a}$ and $\mathbf{b}$:

a $\overrightarrow{PX}$ **b** $\overrightarrow{PS}$ **c** $\overrightarrow{QX}$ **d** $\overrightarrow{RS}$

6 

In the given figure [BD] is parallel to [OA] and half its length. Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, vector expressions for:

a $\overrightarrow{BD}$ **b** $\overrightarrow{AB}$ **c** $\overrightarrow{BA}$
d $\overrightarrow{OD}$ **e** $\overrightarrow{AD}$ **f** $\overrightarrow{DA}$

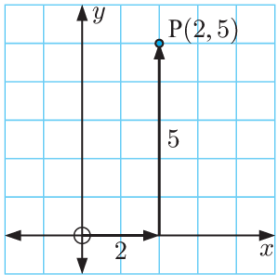
DISCUSSION

- What difficulties are there in using scale diagrams to solve vector problems?
- Can you extend your vector diagrams to higher dimensions?
- Can you draw the zero vector $\mathbf{0}$?

C VECTORS IN THE PLANE

When we plot points in the Cartesian plane, we move first in the x-direction and then in the y-direction.

For example, to plot the point P$(2, 5)$, we start at the origin, move 2 units in the x-direction, and then 5 units in the y-direction.

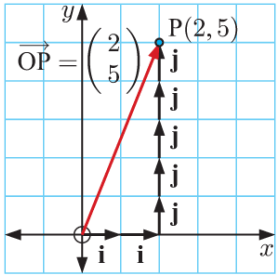


We have previously seen in transformation geometry that translating a point a units in the x-direction and b units in the y-direction can be achieved using the *translation vector* $\begin{pmatrix} a \\ b \end{pmatrix}$.

So, the vector from O to P is $\overrightarrow{OP} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$.

Let $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ be a translation 1 unit in the positive x-direction

and $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ be a translation 1 unit in the positive y-direction.



Moving from O to P is equivalent to two lots of $\mathbf{i}$ plus 5 lots of $\mathbf{j}$.

$$\overrightarrow{OP} = 2\mathbf{i} + 5\mathbf{j}$$

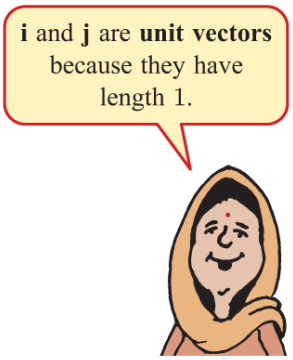
$$\therefore \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 5\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In the 2-dimensional plane, the point $P(x, y)$ has **position vector**

$$\overrightarrow{OP} = \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\text{component form}} = \underbrace{x\mathbf{i} + y\mathbf{j}}_{\text{unit vector form}}$$

where $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is the **base unit vector** in the x-direction

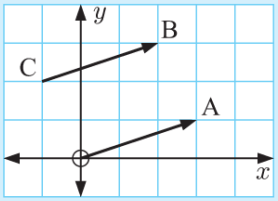
and $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is the **base unit vector** in the y-direction.



VECTOR EQUALITY

Two vectors are **equal** if their components are equal.

Example 11 **Self Tutor**



a Write vectors $\overrightarrow{OA}$ and $\overrightarrow{CB}$ in component form and in unit vector form.

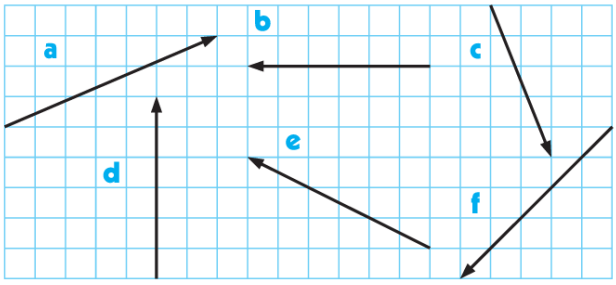
b Comment on your answers in **a**.

a $\overrightarrow{OA} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3\mathbf{i} + \mathbf{j}$ $\qquad$ $\overrightarrow{CB} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3\mathbf{i} + \mathbf{j}$

b The vectors $\overrightarrow{OA}$ and $\overrightarrow{CB}$ are equal.

EXERCISE 12C

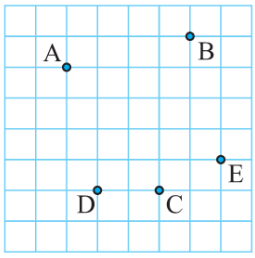
1 Write the illustrated vectors in component form and in unit vector form:



2 Write each vector in unit vector form, and illustrate it using an arrow diagram:

a $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ $\qquad$ **b** $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ $\qquad$ **c** $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ $\qquad$ **d** $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

3



a Write in component form and in unit vector form:

i $\overrightarrow{BA}$ $\qquad$ **ii** $\overrightarrow{BC}$ $\qquad$ **iii** $\overrightarrow{DC}$

iv $\overrightarrow{AC}$ $\qquad$ **v** $\overrightarrow{CA}$ $\qquad$ **vi** $\overrightarrow{DB}$

vii $\overrightarrow{AE}$ $\qquad$ **viii** $\overrightarrow{CE}$ $\qquad$ **ix** $\overrightarrow{ED}$

b Use the given points to describe as many pairs of equal vectors as you can.

4 Write in component form, and illustrate using a directed line segment:

a $\mathbf{i} + 2\mathbf{j}$ **b** $-\mathbf{i} + 3\mathbf{j}$ **c** $-5\mathbf{j}$ **d** $4\mathbf{i} - 2\mathbf{j}$

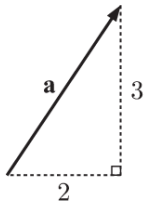
5 Write the zero vector $\mathbf{0}$ in component form.

6 Find the unknowns if:

a $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **b** $\begin{pmatrix} a+1 \\ 2b-8 \end{pmatrix} = \begin{pmatrix} 9-a \\ a \end{pmatrix}$ **c** $\begin{pmatrix} 2x+3y \\ x-2 \end{pmatrix} = \begin{pmatrix} 11 \\ 2y \end{pmatrix}$

D THE MAGNITUDE OF A VECTOR

Consider vector $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 2\mathbf{i} + 3\mathbf{j}$.



The **magnitude** or **length** of $\mathbf{a}$ is represented by $|\mathbf{a}|$.

By Pythagoras, $|\mathbf{a}|^2 = 2^2 + 3^2 = 4 + 9 = 13$

$\therefore |\mathbf{a}| = \sqrt{13}$ units $\quad \{\text{since } |\mathbf{a}| > 0\}$

The **magnitude** or **length** of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is $|\mathbf{a}| = \sqrt{{a_1}^2 + {a_2}^2}$.

Example 12 **Self Tutor**

If $\mathbf{p} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and $\mathbf{q} = 2\mathbf{i} - 5\mathbf{j}$ find: **a** $|\mathbf{p}|$ **b** $|\mathbf{q}|$

a $\mathbf{p} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$

$\therefore |\mathbf{p}| = \sqrt{3^2 + (-5)^2}$

$= \sqrt{34}$ units

b As $2\mathbf{i} - 5\mathbf{j} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$,

$|\mathbf{q}| = \sqrt{2^2 + (-5)^2}$

$= \sqrt{29}$ units

UNIT VECTORS

A **unit vector** is any vector which has a length of one unit.

For example, we have seen that $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are the base unit vectors in the positive x and y-directions respectively.

Example 13 **Self Tutor**

Find k given that $\begin{pmatrix} -\frac{1}{3} \\ k \end{pmatrix}$ is a unit vector.

Since $\begin{pmatrix} -\frac{1}{3} \\ k \end{pmatrix}$ is a unit vector, $\sqrt{(-\frac{1}{3})^2 + k^2} = 1$

$\therefore \sqrt{\frac{1}{9} + k^2} = 1$

$\therefore \frac{1}{9} + k^2 = 1$

$\therefore k^2 = \frac{8}{9}$

$\therefore k = \pm\frac{\sqrt{8}}{3}$

EXERCISE 12D

1 Find the magnitude of:

a $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ **b** $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ **c** $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ **d** $\begin{pmatrix} -2 \\ 2 \end{pmatrix}$ **e** $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

2 Find the length of:

a $\mathbf{i} + \mathbf{j}$ **b** $5\mathbf{i} - 12\mathbf{j}$ **c** $-\mathbf{i} + 4\mathbf{j}$ **d** $3\mathbf{i}$ **e** $k\mathbf{j}$

3 Which of the following are unit vectors?

a $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$ **b** $\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ **c** $\begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$ **d** $\begin{pmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{pmatrix}$ **e** $\begin{pmatrix} \frac{2}{7} \\ -\frac{5}{7} \end{pmatrix}$

4 Find k such that the given vector is a unit vector:

a $\begin{pmatrix} 0 \\ k \end{pmatrix}$ **b** $\begin{pmatrix} k \\ 0 \end{pmatrix}$ **c** $\begin{pmatrix} k \\ 1 \end{pmatrix}$ **d** $\begin{pmatrix} k \\ k \end{pmatrix}$ **e** $\begin{pmatrix} \frac{1}{2} \\ k \end{pmatrix}$

5 Given $\mathbf{v} = \begin{pmatrix} 8 \\ p \end{pmatrix}$ and $|\mathbf{v}| = \sqrt{73}$ units, find the possible values of p.

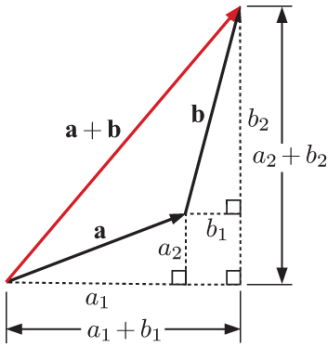
E OPERATIONS WITH PLANE VECTORS

ALGEBRAIC VECTOR ADDITION

Consider adding vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

Notice that:

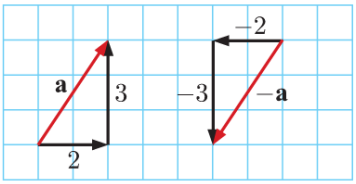
- the horizontal step for $\mathbf{a} + \mathbf{b}$ is $a_1 + b_1$
- the vertical step for $\mathbf{a} + \mathbf{b}$ is $a_2 + b_2$.



If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ then $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$.

ALGEBRAIC NEGATIVE VECTORS

In the diagram we see the vector $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and its negative $-\mathbf{a} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$.



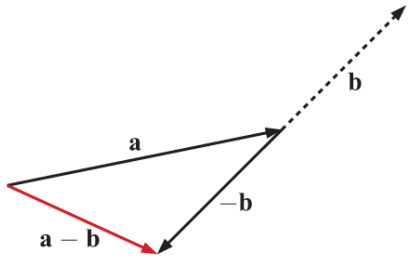
If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ then $-\mathbf{a} = \begin{pmatrix} -a_1 \\ -a_2 \end{pmatrix}$.

ALGEBRAIC VECTOR SUBTRACTION

To subtract one vector from another, we simply **add its negative**.

Suppose $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

$$\therefore \ \mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$$
$$= \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} -b_1 \\ -b_2 \end{pmatrix}$$
$$= \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}$$



If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, then $\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}$.

Example 14 **Self Tutor**

Given $\mathbf{p} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$, find:

a $\mathbf{p} + \mathbf{r}$ **b** $\mathbf{q} - \mathbf{p}$ **c** $\mathbf{p} - \mathbf{q} - \mathbf{r}$

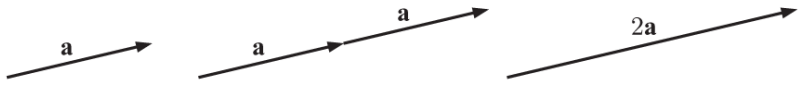
a
$$\mathbf{p} + \mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 3 + (-2) \\ -2 + (-5) \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \end{pmatrix}$$

b
$$\mathbf{q} - \mathbf{p} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 - 3 \\ 4 - (-2) \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$$

c
$$\mathbf{p} - \mathbf{q} - \mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 3 - 1 - (-2) \\ -2 - 4 - (-5) \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

ALGEBRAIC SCALAR MULTIPLICATION

We have already seen a geometric approach for integer scalar multiplication:



Consider $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

$$2\mathbf{a} = \mathbf{a} + \mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \times 1 \\ 2 \times 3 \end{pmatrix}$$
$$3\mathbf{a} = \mathbf{a} + \mathbf{a} + \mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \end{pmatrix} = \begin{pmatrix} 3 \times 1 \\ 3 \times 3 \end{pmatrix}$$

If k is any scalar and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, then $k\mathbf{v} = \begin{pmatrix} kv_1 \\ kv_2 \end{pmatrix}$.

Notice that:

- $(-1)\mathbf{v} = \begin{pmatrix} (-1)\,v_1 \\ (-1)\,v_2 \end{pmatrix} = \begin{pmatrix} -v_1 \\ -v_2 \end{pmatrix} = -\mathbf{v}$
- $(0)\mathbf{v} = \begin{pmatrix} (0)\,v_1 \\ (0)\,v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{0}$

Example 15 Self Tutor

For $\mathbf{p} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ find:

a $3\mathbf{q}$ **b** $\mathbf{p} + 2\mathbf{q}$ **c** $\frac{1}{2}\mathbf{p} - 3\mathbf{q}$

a
$$3\mathbf{q} = 3\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 6 \\ -9 \end{pmatrix}$$

b
$$\mathbf{p} + 2\mathbf{q} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 + 2(2) \\ 1 + 2(-3) \end{pmatrix} = \begin{pmatrix} 8 \\ -5 \end{pmatrix}$$

c
$$\frac{1}{2}\mathbf{p} - 3\mathbf{q} = \frac{1}{2}\begin{pmatrix} 4 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(4) - 3(2) \\ \frac{1}{2}(1) - 3(-3) \end{pmatrix} = \begin{pmatrix} -4 \\ 9\frac{1}{2} \end{pmatrix}$$

EXERCISE 12E

1 If $\mathbf{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$ find:

a $\mathbf{a} + \mathbf{b}$ **b** $\mathbf{b} + \mathbf{a}$ **c** $\mathbf{b} + \mathbf{c}$ **d** $\mathbf{c} + \mathbf{b}$

e $\mathbf{a} + \mathbf{c}$ **f** $\mathbf{c} + \mathbf{a}$ **g** $\mathbf{a} + \mathbf{a}$ **h** $\mathbf{b} + \mathbf{a} + \mathbf{c}$

2 Given $\mathbf{p} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ find:

a $\mathbf{p} - \mathbf{q}$ **b** $\mathbf{q} - \mathbf{r}$ **c** $\mathbf{p} + \mathbf{q} - \mathbf{r}$

d $\mathbf{p} - \mathbf{q} - \mathbf{r}$ **e** $\mathbf{q} - \mathbf{r} - \mathbf{p}$ **f** $\mathbf{r} + \mathbf{q} - \mathbf{p}$

3 Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

a Use vector addition to show that $\mathbf{a} + \mathbf{0} = \mathbf{a}$.

b Use vector subtraction to show that $\mathbf{a} - \mathbf{a} = \mathbf{0}$.

4 For $\mathbf{p} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$ find:

a $-3\mathbf{p}$ **b** $\frac{1}{2}\mathbf{q}$ **c** $2\mathbf{p} + \mathbf{q}$ **d** $\mathbf{p} - 2\mathbf{q}$

e $\mathbf{p} - \frac{1}{2}\mathbf{r}$ **f** $2\mathbf{p} + 3\mathbf{r}$ **g** $2\mathbf{q} - 3\mathbf{r}$ **h** $2\mathbf{p} - \mathbf{q} + \frac{1}{3}\mathbf{r}$

5 Consider $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

a Find geometrically:

i $\mathbf{p} + \mathbf{p} + \mathbf{q} + \mathbf{q} + \mathbf{q}$ **ii** $\mathbf{p} + \mathbf{q} + \mathbf{p} + \mathbf{q} + \mathbf{q}$ **iii** $\mathbf{q} + \mathbf{p} + \mathbf{q} + \mathbf{p} + \mathbf{q}$

b Comment on your results in **a**.

6 Find k such that:

a $\begin{pmatrix} 2 \\ -5 \end{pmatrix} + k\begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 20 \\ -11 \end{pmatrix}$ **b** $\frac{1}{3}\begin{pmatrix} -1 \\ k \end{pmatrix} + \frac{4}{3}\begin{pmatrix} 7 \\ k+1 \end{pmatrix} = \begin{pmatrix} 9 \\ -1 \end{pmatrix}$

Example 16 **Self Tutor**

If $\mathbf{p} = 3\mathbf{i} - 5\mathbf{j}$ and $\mathbf{q} = -\mathbf{i} - 2\mathbf{j}$, find $|\mathbf{p} - 2\mathbf{q}|$.

$$\begin{aligned}\mathbf{p} - 2\mathbf{q} &= 3\mathbf{i} - 5\mathbf{j} - 2(-\mathbf{i} - 2\mathbf{j}) \\ &= 3\mathbf{i} - 5\mathbf{j} + 2\mathbf{i} + 4\mathbf{j} \\ &= 5\mathbf{i} - \mathbf{j}\end{aligned}$$

$$\therefore\ |\mathbf{p} - 2\mathbf{q}| = \sqrt{5^2 + (-1)^2} = \sqrt{26} \text{ units}$$

7 If $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{s} = -\mathbf{i} + 4\mathbf{j}$, find:

a $|\mathbf{r}|$ **b** $|\mathbf{s}|$ **c** $|\mathbf{r} + \mathbf{s}|$ **d** $|\mathbf{r} - \mathbf{s}|$

e $|\mathbf{s} - 2\mathbf{r}|$ **f** $|2\mathbf{r}|$ **g** $|\mathbf{r} + 2\mathbf{s}|$ **h** $|2\mathbf{r} - \mathbf{s}|$

8 **a** If $\mathbf{p} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, find:

i $|\mathbf{p}|$ **ii** $|2\mathbf{p}|$ **iii** $|-2\mathbf{p}|$ **iv** $|3\mathbf{p}|$ **v** $|-3\mathbf{p}|$

b By letting $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, prove that $|k\mathbf{v}| = |k|\,|\mathbf{v}|$.

9 Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

Show by equating components, that if $k\mathbf{x} = \mathbf{a}$ then $\mathbf{x} = \frac{1}{k}\mathbf{a}$.

ACTIVITY **SKI RACER**

Click on the icon to practise your skills with vectors.



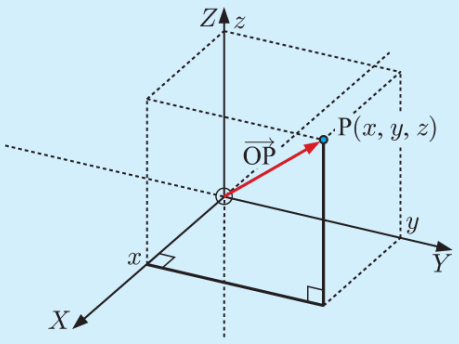
F VECTORS IN SPACE

In **Chapter 10** of the **Core Topics HL** book, we saw that we can specify a point in 3-dimensional space using 3 mutually perpendicular axes called the X, Y, and Z axes.

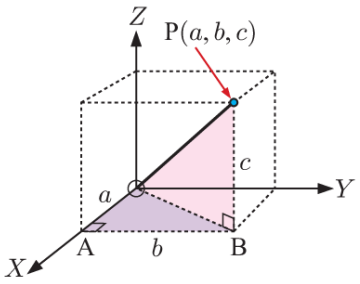
In 3-dimensional space, the **position vector** of $\mathrm{P}(x, y, z)$ is $\overrightarrow{\mathrm{OP}} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

where $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

are the **base unit vectors** in the X, Y, and Z directions respectively.



THE MAGNITUDE OF A VECTOR



Triangle OAB is right angled at A

$\therefore \ OB^2 = a^2 + b^2 \quad .... \ (1)$ {Pythagoras}

Triangle OBP is right angled at B

$\therefore \ OP^2 = OB^2 + c^2$ {Pythagoras}

$\therefore \ OP^2 = a^2 + b^2 + c^2$ {using (1)}

$\therefore \ OP = \sqrt{a^2 + b^2 + c^2}$

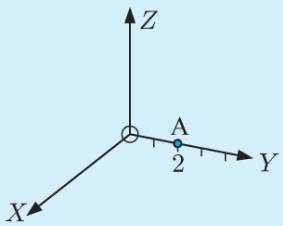
The magnitude or length of the vector $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ is $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

Example 17 **Self Tutor**

Illustrate each point and find its distance from the origin O:

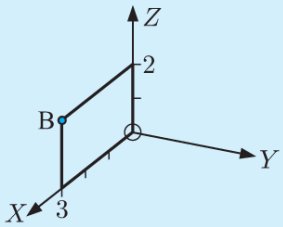
a A(0, 2, 0) **b** B(3, 0, 2) **c** C(−1, 2, 3)

a



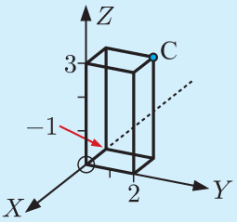
$|\overrightarrow{OA}|$
$= \sqrt{0^2 + 2^2 + 0^2}$
$= 2$ units

b



$|\overrightarrow{OB}|$
$= \sqrt{3^2 + 0^2 + 2^2}$
$= \sqrt{13}$ units

c



$|\overrightarrow{OC}|$
$= \sqrt{(-1)^2 + 2^2 + 3^2}$
$= \sqrt{14}$ units

EXERCISE 12F

1 Consider the point T(3, −1, 4).

a Draw a diagram to locate the position of T in space. **b** Find $\overrightarrow{OT}$.

c Find the distance from O to T.

2 For each point P, write $\overrightarrow{OP}$ in component form and unit vector form:

a P(2, 0, 4) **b** P(−3, 1, 2) **c** P(0, −2, 0)

3 Find the length of:

a $\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$ **b** $\begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix}$ **c** $\begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix}$ **d** $\begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$

4 Illustrate P and find its distance from the origin O:

a P(0, 0, −3) **b** P(0, −1, 2) **c** P(3, 1, 4) **d** P(−1, −2, 3)

5 Write each vector in component form, and find its length:

a $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$ **b** $\mathbf{b} = -3\mathbf{i} + 4\mathbf{k}$ **c** $\mathbf{c} = 2\mathbf{j} - 3\mathbf{k}$

6 Find a, b, and c if:

a $\begin{pmatrix} a-4 \\ b-3 \\ c+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$ **b** $\begin{pmatrix} a-5 \\ b-2 \\ c+3 \end{pmatrix} = \begin{pmatrix} 3-a \\ 2-b \\ 5-c \end{pmatrix}$ **c** $\begin{pmatrix} 3a+4 \\ b-c \\ 4-5b \end{pmatrix} = \begin{pmatrix} 12-a \\ a-4 \\ 5a-22 \end{pmatrix}$

7 Which of the following are unit vectors?

a $\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ **b** $\begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$ **c** $\begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$ **d** $\begin{pmatrix} \frac{1}{3\sqrt{3}} \\ \frac{5}{3\sqrt{3}} \\ -\frac{1}{3\sqrt{3}} \end{pmatrix}$



8 Find k given the unit vector:

a $\begin{pmatrix} -\frac{1}{2} \\ k \\ \frac{1}{4} \end{pmatrix}$ **b** $\begin{pmatrix} k \\ \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}$ **c** $\begin{pmatrix} k \\ k \\ k \end{pmatrix}$ **d** $\begin{pmatrix} \frac{1}{k} \\ \frac{2}{k} \\ \frac{3}{k} \end{pmatrix}$

9 Find the possible values of m, given that:

a $\begin{pmatrix} -2 \\ m \\ 1 \end{pmatrix}$ has length $\sqrt{14}$ units **b** $\begin{pmatrix} 4 \\ -3 \\ m \end{pmatrix}$ has length 6 units.

10 Suppose $\left|\begin{pmatrix} a \\ b \end{pmatrix}\right| = 5$ units and $\left|\begin{pmatrix} b \\ c \end{pmatrix}\right| = 3$ units. Find the range of possible values of $\left|\begin{pmatrix} a \\ b \\ c \end{pmatrix}\right|$.

G OPERATIONS WITH VECTORS IN SPACE

For 3-dimensional vectors, it is difficult to illustrate operations. Instead, we perform operations using component form.

The rules for algebra with vectors readily extend from 2 dimensions to 3 dimensions:

If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ then:

- $-\mathbf{a} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$
- $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$
- $\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{pmatrix}$
- $k\mathbf{a} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}$ for any scalar k.

Example 18 Self Tutor

If $\mathbf{a} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$, find:

a $\mathbf{a} + \mathbf{b}$ **b** $2\mathbf{a} - \mathbf{b}$ **c** $|\mathbf{a}|\,\mathbf{b}$

a
$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1+3 \\ 3+0 \\ 2+(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

b
$$2\mathbf{a} - \mathbf{b} = 2\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2-3 \\ 6-0 \\ 4-(-1) \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \\ 5 \end{pmatrix}$$

c
$$|\mathbf{a}|\,\mathbf{b} = \sqrt{(-1)^2+3^2+2^2}\begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \sqrt{14}\begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3\sqrt{14} \\ 0 \\ -\sqrt{14} \end{pmatrix}$$

EXERCISE 12G

1 Let $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$. Find:

a $\mathbf{a} + \mathbf{b}$ **b** $\mathbf{a} - \mathbf{b}$ **c** $\mathbf{b} + 2\mathbf{c}$

d $\mathbf{c} - \frac{1}{2}\mathbf{a}$ **e** $\mathbf{a} - \mathbf{b} - \mathbf{c}$ **f** $2\mathbf{b} - \mathbf{c} + \mathbf{a}$

2 Let $\mathbf{a} = \mathbf{i} + 3\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + \mathbf{j} + \mathbf{k}$. Find:

a $|\mathbf{a}|$ **b** $|\mathbf{b}|$ **c** $2|\mathbf{a}|$ **d** $|2\mathbf{a}|$

e $-3|\mathbf{b}|$ **f** $|-3\mathbf{b}|$ **g** $|\mathbf{a} + \mathbf{b}|$ **h** $|\mathbf{a} - \mathbf{b}|$

3 Let $\mathbf{a} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} -2 \\ 2 \\ 4 \end{pmatrix}$. Find:

a $2\mathbf{a} - \mathbf{c}$ **b** $3\mathbf{b} + \frac{1}{2}\mathbf{c}$ **c** $-\mathbf{a} + 3\mathbf{b}$ **d** $\mathbf{a} - 5\mathbf{b} + 4\mathbf{c}$

e $|\mathbf{b} + \mathbf{c}|$ **f** $|\mathbf{a} - \mathbf{c}|$ **g** $|\mathbf{a}|\,\mathbf{b}$ **h** $\frac{1}{|\mathbf{a}|}\mathbf{a}$

4 Find scalars a, b, and c such that:

a $2\begin{pmatrix} 1 \\ 0 \\ 3a \end{pmatrix} = \begin{pmatrix} b \\ c-1 \\ 2 \end{pmatrix}$ **b** $a\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + c\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 3 \end{pmatrix}$

c $a\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + b\begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ c \\ 2 \end{pmatrix}$ **d** $2\begin{pmatrix} a \\ 1 \\ -2 \end{pmatrix} + \begin{pmatrix} b \\ c \\ 4 \end{pmatrix} = \begin{pmatrix} c \\ -c \\ ab \end{pmatrix}$

H VECTOR ALGEBRA

INVESTIGATION 1 PROPERTIES OF VECTORS IN SPACE

There are several properties which are valid for the addition of real numbers. For example, we know that $a + b = b + a$. In this Investigation we identify similar properties of vectors.

What to do:

1 Use general vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ to prove that:

a $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ **b** $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$

c $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$ **d** $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$

2 Prove that for scalar k and vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$:

a $|k\mathbf{a}| = |k|\,|\mathbf{a}|$ **b** $k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$

From the **Investigation** you should have found that for vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and scalar $k \in \mathbb{R}$:

- $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ {commutative property}
- $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ {associative property}
- $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$ {additive identity}
- $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$ {additive inverse}
- $k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$ {distributive property}
- $|k\mathbf{a}| = |k|\,|\mathbf{a}|$ where $k\mathbf{a}$ is parallel to $\mathbf{a}$

  $|k\mathbf{a}|$: length of $k\mathbf{a}$; $|k|$: absolute value of k; $|\mathbf{a}|$: length of $\mathbf{a}$

The rules for solving vector equations are similar to those for solving real number equations. However, there is no such thing as dividing a vector by a scalar. Instead, we multiply by the reciprocal scalar.

For example, if $2\mathbf{x} = \mathbf{a}$ then $\mathbf{x} = \frac{1}{2}\mathbf{a}$ and *not* $\frac{\mathbf{a}}{2}$. $\frac{\mathbf{a}}{2}$ has no meaning in vector algebra.

Two useful rules are:

- If $\mathbf{x} + \mathbf{a} = \mathbf{b}$ then $\mathbf{x} = \mathbf{b} - \mathbf{a}$.
- If $k\mathbf{x} = \mathbf{a}$, $k \neq 0$, then $\mathbf{x} = \frac{1}{k}\mathbf{a}$.

Proof:

- If $\mathbf{x} + \mathbf{a} = \mathbf{b}$
  then $\mathbf{x} + \mathbf{a} + (-\mathbf{a}) = \mathbf{b} + (-\mathbf{a})$
  $\therefore\ \mathbf{x} + \mathbf{0} = \mathbf{b} - \mathbf{a}$
  $\therefore\ \mathbf{x} = \mathbf{b} - \mathbf{a}$

- If $k\mathbf{x} = \mathbf{a}$
  then $\frac{1}{k}(k\mathbf{x}) = \frac{1}{k}\mathbf{a}$
  $\therefore\ 1\mathbf{x} = \frac{1}{k}\mathbf{a}$
  $\therefore\ \mathbf{x} = \frac{1}{k}\mathbf{a}$

Example 19 **Self Tutor**

Solve for $\mathbf{x}$:

a $3\mathbf{x} - \mathbf{r} = \mathbf{s}$ **b** $\mathbf{c} - 2\mathbf{x} = \mathbf{d}$

a
$3\mathbf{x} - \mathbf{r} = \mathbf{s}$
$\therefore\ 3\mathbf{x} = \mathbf{s} + \mathbf{r}$
$\therefore\ \mathbf{x} = \frac{1}{3}(\mathbf{s} + \mathbf{r})$

b
$\mathbf{c} - 2\mathbf{x} = \mathbf{d}$
$\therefore\ \mathbf{c} - \mathbf{d} = 2\mathbf{x}$
$\therefore\ \frac{1}{2}(\mathbf{c} - \mathbf{d}) = \mathbf{x}$

EXERCISE 12H

1 Solve the following vector equations for $\mathbf{x}$:

a $2\mathbf{x} = \mathbf{q}$ **b** $\frac{1}{2}\mathbf{x} = \mathbf{n}$ **c** $-3\mathbf{x} = \mathbf{p}$

d $\mathbf{q} + 2\mathbf{x} = \mathbf{r}$ **e** $4\mathbf{s} - 5\mathbf{x} = \mathbf{t}$ **f** $4\mathbf{m} - \frac{1}{3}\mathbf{x} = \mathbf{n}$

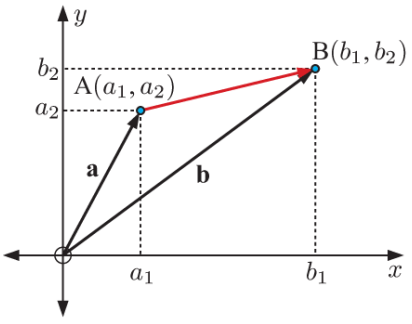
2 Suppose $\mathbf{p} = \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix}$. Find $\mathbf{x}$ such that:

a $2\mathbf{x} = \mathbf{p}$ **b** $\frac{1}{3}\mathbf{x} = \mathbf{p}$ **c** $4\mathbf{x} + \mathbf{p} = \mathbf{0}$

3 Suppose $\mathbf{a} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$. Find $\mathbf{x}$ such that:

a $2\mathbf{a} + \mathbf{x} = \mathbf{b}$ **b** $3\mathbf{x} - \mathbf{a} = 2\mathbf{b}$ **c** $2\mathbf{b} - 2\mathbf{x} = -\mathbf{a}$

I THE VECTOR BETWEEN TWO POINTS

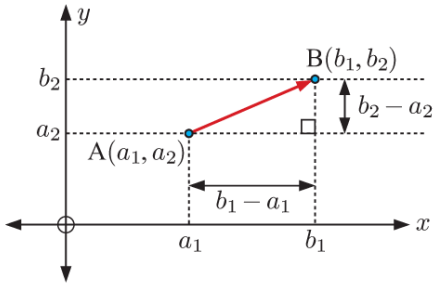


In the diagram, point A has position vector $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, and point B has position vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

$\therefore\ \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$
$= -\mathbf{a} + \mathbf{b}$
$= \mathbf{b} - \mathbf{a}$
$= \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} - \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$
$= \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$

The **displacement vector of B relative to A** is $\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$.

Notice that $\overrightarrow{AB} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$ is the *translation vector* required to move point A to point B.



Example 20 **Self Tutor**

Given points A$(-1, 2)$, B$(3, 4)$, and C$(4, -5)$, find the displacement vector of:

a B from O **b** B from A **c** A from C.

a The displacement vector of B relative to O is $\overrightarrow{OB} = \begin{pmatrix} 3-0 \\ 4-0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

b The displacement vector of B relative to A is $\overrightarrow{AB} = \begin{pmatrix} 3--1 \\ 4-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$.

c The displacement vector of A relative to C is $\overrightarrow{CA} = \begin{pmatrix} -1-4 \\ 2--5 \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$.

We can define a similar result for vectors in space:

If A(a_1, a_2, a_3) and B(b_1, b_2, b_3) are two points in space with position vectors **a** and **b** respectively, then the **displacement vector of B relative to A** is

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \\ b_3 - a_3 \end{pmatrix} \begin{matrix} \leftarrow X\text{-step} \\ \leftarrow Y\text{-step} \\ \leftarrow Z\text{-step} \end{matrix}$$

Example 21 **Self Tutor**

Given P$(-3, 1, 2)$ and Q$(1, -1, 3)$, find: **a** $\overrightarrow{OP}$ **b** $\overrightarrow{PQ}$ **c** $|\overrightarrow{PQ}|$

a $\overrightarrow{OP} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$ **b** $\overrightarrow{PQ} = \begin{pmatrix} 1-(-3) \\ -1-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$

c $|\overrightarrow{PQ}| = \sqrt{4^2 + (-2)^2 + 1^2}$
$= \sqrt{21}$ units

EXERCISE 12I

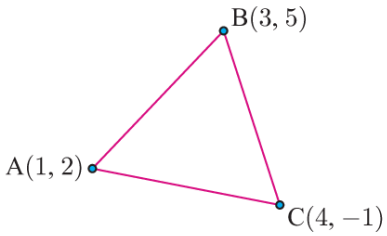
1 Find $\overrightarrow{AB}$ given:

a A$(2, 3)$ and B$(4, 7)$ **b** A$(3, -1)$ and B$(1, 4)$ **c** A$(-2, 7)$ and B$(1, 4)$.

2 Given $A(2, 3)$ and $B(6, 2)$, find:

a $\overrightarrow{AB}$ **b** $\overrightarrow{BA}$ **c** $|\overrightarrow{AB}|$

3



a Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b Explain why $\overrightarrow{BC} = -\overrightarrow{AB} + \overrightarrow{AC}$.

c Hence find $\overrightarrow{BC}$.

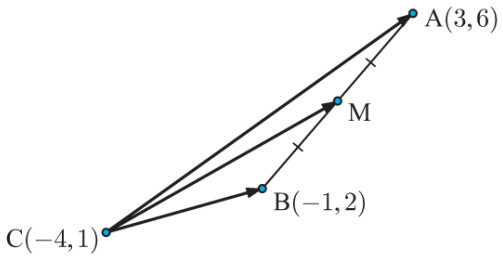
d Check your answer to **c** by direct evaluation.

4 **a** Given $\overrightarrow{BA} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$, find $\overrightarrow{AC}$.

b Given $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ and $\overrightarrow{CA} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, find $\overrightarrow{CB}$.

c Given $\overrightarrow{PQ} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, $\overrightarrow{RQ} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, and $\overrightarrow{RS} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, find $\overrightarrow{SP}$.

5



a Find the coordinates of M.

b Find vectors $\overrightarrow{CA}$, $\overrightarrow{CM}$, and $\overrightarrow{CB}$.

c Verify that $\overrightarrow{CM} = \frac{1}{2}\overrightarrow{CA} + \frac{1}{2}\overrightarrow{CB}$.

6 Find $\overrightarrow{PQ}$ given:

a $P(1, 0, 2)$ and $Q(3, 5, 4)$ **b** $P(5, 2, 3)$ and $Q(6, 2, -1)$

c $P(-2, 3, 0)$ and $Q(1, 4, -3)$ **d** $P(4, -1, 5)$ and $Q(-1, -5, 3)$.

7 If $\overrightarrow{OA} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$, find $\overrightarrow{AB}$ and hence the distance from A to B.

8 Given $A(-3, 1, 2)$ and $B(1, 0, -1)$, find:

a $\overrightarrow{AB}$ and $\overrightarrow{BA}$ **b** the lengths $|\overrightarrow{AB}|$ and $|\overrightarrow{BA}|$.

9 Given $M(4, -2, -1)$ and $N(-1, 2, 0)$, find:

a the displacement vector of M relative to N **b** the displacement vector of N relative to M

c the distance between M and N.

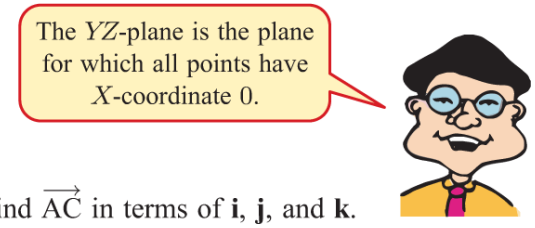
10 For $A(-1, 3, -2)$ and $B(3, -2, 1)$ find:

a $\overrightarrow{AB}$ in terms of **i**, **j**, and **k** **b** the magnitude of $\overrightarrow{AB}$.

11 Find the shortest distance from $Q(3, 1, -2)$ to:

a the Y-axis **b** the origin

c the YZ-plane.



12 If $\overrightarrow{AB} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\overrightarrow{BC} = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, find $\overrightarrow{AC}$ in terms of **i**, **j**, and **k**.

13 If $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$, and $\overrightarrow{BD} = \begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix}$, find:

a $\overrightarrow{AD}$ **b** $\overrightarrow{CB}$ **c** $\overrightarrow{CD}$

14 Consider the points $A(-1, 3)$ and $B(3, k)$.

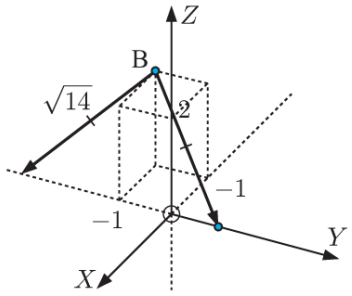
a Find $\overrightarrow{AB}$ and $|\overrightarrow{AB}|$.

b Given that A and B are 5 units apart, find the two possible values of k.

c Show, by illustration, why k should have two possible values.

15 **a** State the coordinates of any general point A on the Y-axis.

b Hence find the coordinates of two points on the Y-axis which are $\sqrt{14}$ units from $B(-1, -1, 2)$.



Example 22 **Self Tutor**

[AB] is the diameter of a circle with centre $C(-1, 2)$. If B is $(3, 1)$, find:

a $\overrightarrow{BC}$ **b** the coordinates of A.

a $\overrightarrow{BC} = \begin{pmatrix} -1-3 \\ 2-1 \end{pmatrix} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$

b If A has coordinates (a, b), then

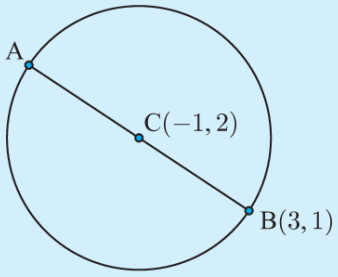
$\overrightarrow{CA} = \begin{pmatrix} a-(-1) \\ b-2 \end{pmatrix} = \begin{pmatrix} a+1 \\ b-2 \end{pmatrix}$

But $\overrightarrow{CA} = \overrightarrow{BC}$, so $\begin{pmatrix} a+1 \\ b-2 \end{pmatrix} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$\therefore\ a+1 = -4$ and $b-2 = 1$

$\therefore\ a = -5$ and $b = 3$

$\therefore$ A is $(-5, 3)$.



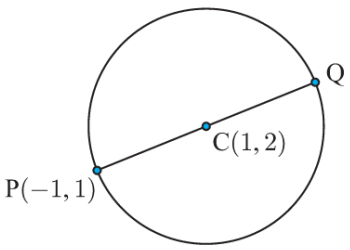
16 Consider the point $A(1, 4)$. Find the coordinates of:

a B given $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ **b** C given $\overrightarrow{CA} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

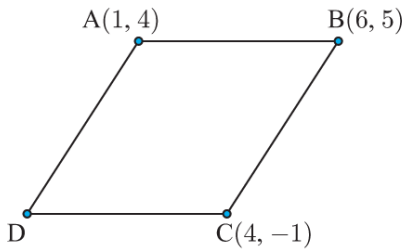
17 [PQ] is the diameter of a circle with centre C.

a Find $\overrightarrow{PC}$.

b Hence find the coordinates of Q.



18



ABCD is a parallelogram.

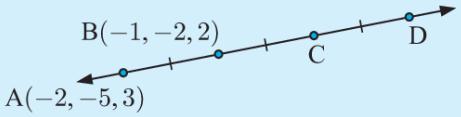
a Find $\overrightarrow{AB}$.

b Find $\overrightarrow{CD}$.

c Hence find the coordinates of D.

Example 23 **Self Tutor**

Find the coordinates of C and D:



$$\overrightarrow{AB} = \begin{pmatrix} -1-(-2) \\ -2-(-5) \\ 2-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$$
$$= \overrightarrow{OA} + 2\overrightarrow{AB}$$
$$= \begin{pmatrix} -2 \\ -5 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 6 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

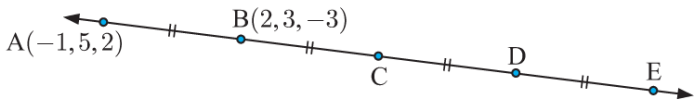
$\therefore$ C is $(0, 1, 1)$

$$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$$
$$= \overrightarrow{OA} + 3\overrightarrow{AB}$$
$$= \begin{pmatrix} -2 \\ -5 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 9 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}$$

$\therefore$ D is $(1, 4, 0)$

19 Consider the points $A(2, 1, -2)$, $B(0, 3, -4)$, $C(1, -2, 1)$, and $D(-2, -3, 2)$.
Show that $\overrightarrow{BD} = 2\overrightarrow{AC}$.

20 Find the coordinates of C, D, and E.



21 A sphere has centre $C(-1, 2, 4)$ and diameter [AB] where A is $(-2, 1, 3)$.
Find the coordinates of B and the radius of the sphere.

22 Consider $A(-1, 2, 5)$, $B(2, 0, 3)$, and $C(-3, 1, 0)$.

a Find the displacement vector $\overrightarrow{AB}$ and its length AB.

b Find the displacement vector $\overrightarrow{AC}$ and its length AC.

c Find the displacement vector $\overrightarrow{CB}$ and its length CB.

d Hence classify triangle ABC.

23 Show that $P(0, 4, 4)$, $Q(2, 6, 5)$, and $R(1, 4, 3)$ are vertices of an isosceles triangle.

24 Use side lengths to classify triangle ABC given the coordinates:

a $A(0, 0, 3)$, $B(2, 8, 1)$, and $C(-9, 6, 18)$

b $A(1, 0, -3)$, $B(2, 2, 0)$, and $C(4, 6, 6)$.

25 The vertices of triangle ABC are $A(5, 6, -2)$, $B(6, 12, 9)$, and $C(2, 4, 2)$.

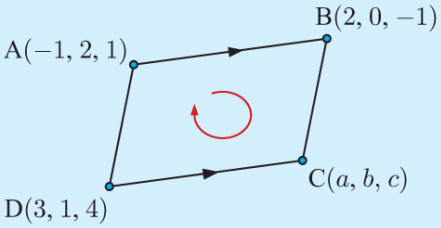
a Show that the triangle is right angled. **b** Hence find the area of the triangle.

Example 24 **Self Tutor**

ABCD is a parallelogram. A is $(-1, 2, 1)$, B is $(2, 0, -1)$, and D is $(3, 1, 4)$. Find the coordinates of C.

Let C be (a, b, c).

[AB] is parallel to [DC], and they have the same length, so $\overrightarrow{DC} = \overrightarrow{AB}$



$$\therefore \begin{pmatrix} a-3 \\ b-1 \\ c-4 \end{pmatrix} = \begin{pmatrix} 2-(-1) \\ 0-2 \\ -1-1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix}$$

$\therefore\ a-3=3, \quad b-1=-2, \quad$ and $\quad c-4=-2$

$\therefore\ a=6, \quad b=-1, \quad$ and $\quad c=2$

So, C is $(6, -1, 2)$.

Check: The midpoint of [DB] is $\left(\frac{3+2}{2}, \frac{1+0}{2}, \frac{4+-1}{2}\right)$ which is $\left(\frac{5}{2}, \frac{1}{2}, \frac{3}{2}\right)$.

The midpoint of [AC] is $\left(\frac{-1+6}{2}, \frac{2+-1}{2}, \frac{1+2}{2}\right)$ which is $\left(\frac{5}{2}, \frac{1}{2}, \frac{3}{2}\right)$.

The midpoints are the same, so the diagonals of the parallelogram bisect. ✓

26 $A(-1, 3, 4)$, $B(2, 5, -1)$, $C(-1, 2, -2)$, and $D(r, s, t)$ are four points in space. Find r, s, and t if:

a $\overrightarrow{AC} = \overrightarrow{BD}$ **b** $\overrightarrow{AB} = \overrightarrow{DC}$

27 A quadrilateral has vertices $A(1, 2, 3)$, $B(3, -3, 2)$, $C(7, -4, 5)$, and $D(5, 1, 6)$.

a Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$.

b What can be deduced about the quadrilateral ABCD?

28 PQRS is a parallelogram. P is $(-1, 2, 3)$, Q is $(1, -2, 5)$, and R is $(0, 4, -1)$.

a Use vectors to find the coordinates of S.

b Use the midpoints of the diagonals to check your answer.

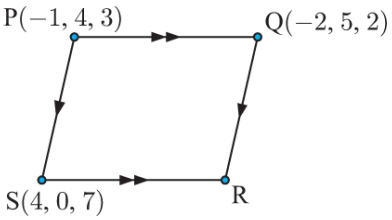
29 Use vectors to determine whether ABCD is a parallelogram:

a $A(5, 0, 3)$, $B(-1, 2, 4)$, $C(4, -3, 6)$, and $D(10, -5, 5)$

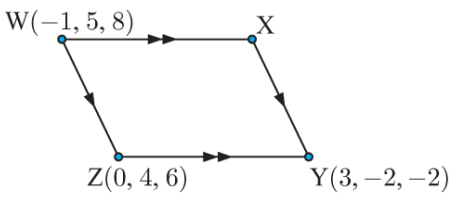
b $A(2, -3, 2)$, $B(1, 4, -1)$, $C(-2, 6, -2)$, and $D(-1, -1, 2)$.

30 Use vector methods to find the remaining vertex of:

a



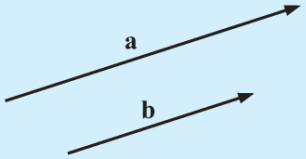
b



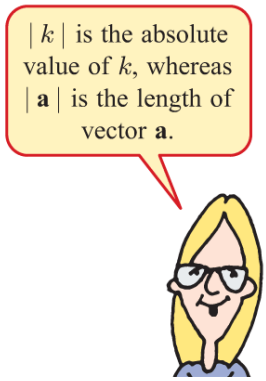
J PARALLELISM

Two non-zero vectors are **parallel** if and only if one is a scalar multiple of the other.

Given any non-zero vector $\mathbf{v}$ and non-zero scalar k, the vector $k\mathbf{v}$ is parallel to $\mathbf{v}$.



- If $\mathbf{a}$ is parallel to $\mathbf{b}$, then there exists a scalar k such that $\mathbf{a} = k\mathbf{b}$.
- If $\mathbf{a} = k\mathbf{b}$ for some scalar k, then
  - $\mathbf{a}$ is parallel to $\mathbf{b}$, and
  - $|\mathbf{a}| = |k|\,|\mathbf{b}|$.



Example 25 **Self Tutor**

Find r and s given that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ r \end{pmatrix}$ is parallel to $\mathbf{b} = \begin{pmatrix} s \\ 2 \\ -3 \end{pmatrix}$.

Since $\mathbf{a}$ and $\mathbf{b}$ are parallel, $\mathbf{a} = k\mathbf{b}$ for some scalar k.

$$\therefore \begin{pmatrix} 2 \\ -1 \\ r \end{pmatrix} = k \begin{pmatrix} s \\ 2 \\ -3 \end{pmatrix}$$

$\therefore\ 2 = ks,\ -1 = 2k,$ and $r = -3k$

Consequently, $k = -\frac{1}{2}$

$\therefore\ 2 = -\frac{1}{2}s$ and $r = -3(-\frac{1}{2})$

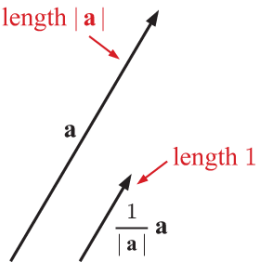
$\therefore\ s = -4$ and $r = \frac{3}{2}$

UNIT VECTORS

Given a non-zero vector $\mathbf{a}$, its magnitude $|\mathbf{a}|$ is a scalar quantity.

If we multiply $\mathbf{a}$ by the scalar $\frac{1}{|\mathbf{a}|}$, we obtain the parallel vector $\frac{1}{|\mathbf{a}|}\mathbf{a}$.

The length of this vector is $\left|\frac{1}{|\mathbf{a}|}\right| |\mathbf{a}| = \frac{|\mathbf{a}|}{|\mathbf{a}|} = 1$, so $\frac{1}{|\mathbf{a}|}\mathbf{a}$ is a unit vector in the direction of $\mathbf{a}$. We write this unit vector as $\hat{\mathbf{a}}$.



- A unit vector in the direction of $\mathbf{a}$ is $\hat{\mathbf{a}} = \frac{1}{|\mathbf{a}|}\mathbf{a}$.
- A vector $\mathbf{b}$ of length k in the same direction as $\mathbf{a}$ is $\mathbf{b} = k\hat{\mathbf{a}} = \frac{k}{|\mathbf{a}|}\mathbf{a}$.
- A vector $\mathbf{b}$ of length k which is *parallel to* $\mathbf{a}$ could be $\mathbf{b} = \pm k\hat{\mathbf{a}} = \pm\frac{k}{|\mathbf{a}|}\mathbf{a}$.

Example 26 **Self Tutor**

Find a vector $\mathbf{b}$ of length 7 units in the opposite direction to $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

The unit vector in the direction of $\mathbf{a}$ is

$$\frac{1}{|\mathbf{a}|}\mathbf{a} = \frac{1}{\sqrt{4+1+1}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

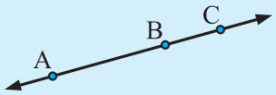
$$\therefore \quad \mathbf{b} = -\frac{7}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$



COLLINEAR POINTS

Three or more points are said to be **collinear** if they lie on the same straight line.

A, B, and C are **collinear** if $\overrightarrow{AB} = k\overrightarrow{BC}$ for some scalar k.



Example 27 **Self Tutor**

Prove that A$(-1, 2, 3)$, B$(4, 0, -1)$, and C$(14, -4, -9)$ are collinear.

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ -2 \\ -4 \end{pmatrix} \text{ and } \overrightarrow{BC} = \begin{pmatrix} 10 \\ -4 \\ -8 \end{pmatrix} = 2\begin{pmatrix} 5 \\ -2 \\ -4 \end{pmatrix}$$

$\therefore \quad \overrightarrow{BC} = 2\overrightarrow{AB}$

$\therefore$ [BC] is parallel to [AB].

Since B is common to both, A, B, and C are collinear.

EXERCISE 12J

1 Determine whether each pair of vectors is parallel:

a $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 15 \end{pmatrix}$ **b** $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -10 \end{pmatrix}$ **c** $\begin{pmatrix} 6 \\ -12 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 8 \end{pmatrix}$

d $\begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ -8 \\ 6 \end{pmatrix}$ **e** $\begin{pmatrix} 8 \\ 0 \\ -20 \end{pmatrix}$ and $\begin{pmatrix} -6 \\ 0 \\ 15 \end{pmatrix}$ **f** $\begin{pmatrix} 6 \\ 10 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 24 \\ 40 \\ 8 \end{pmatrix}$

2 Find a such that:

a $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -6 \\ a \end{pmatrix}$ are parallel **b** $\begin{pmatrix} a \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ are parallel.

3 Find r and s given that $\mathbf{a}$ and $\mathbf{b}$ are parallel:

a $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ r \\ s \end{pmatrix}$ **b** $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} r \\ 2 \\ s \end{pmatrix}$

4 What can be deduced from the following?

a $\overrightarrow{AB} = 3\overrightarrow{CD}$ **b** $\overrightarrow{RS} = -\frac{1}{2}\overrightarrow{KL}$ **c** $\overrightarrow{AB} = 2\overrightarrow{BC}$

5 The position vectors of P, Q, R, and S are $\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$, $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, and $\begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix}$ respectively.

Show that [PR] and [QS] are parallel, and state the ratio PR : QS.

6 If $\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, write down the vector:

a in the same direction as $\mathbf{a}$ and with twice its length

b in the opposite direction to $\mathbf{a}$ and with half its length.

7 Find the 2-dimensional unit vector in the direction:

a $\mathbf{i} + 2\mathbf{j}$ **b** $2\mathbf{i} - 3\mathbf{j}$ **c** $-2\mathbf{i} + 2\mathbf{j}$

8 Find the 3-dimensional unit vector in the direction:

a $2\mathbf{i} + 3\mathbf{k}$ **b** $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ **c** $2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

9 Find a vector $\mathbf{v}$ which has:

a the same direction as $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and with length 3 units

b the opposite direction to $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ and with length 2 units

c the same direction as $\begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$ and with length 6 units

d the opposite direction to $-\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and with length 5 units.

10 **a** Find the vectors of length 1 unit which are parallel to $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$.

b Find vectors of length 2 units which are parallel to $\mathbf{b} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$.

11 A is $(3, 2)$ and point B is 4 units from A in the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

a Find $\overrightarrow{AB}$. **b** Find $\overrightarrow{OB}$ using $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$.

c Hence deduce the coordinates of B.

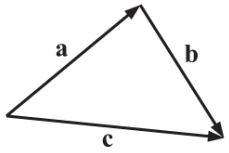
12 Find the coordinates of point X which is:

a 5 units from $(1, -5)$ in the direction $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

b 6 units from $(1, 3, -2)$ in the direction $\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}$

c 4 units from $(2, -1, 4)$ in the direction $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$.

13 The **triangle inequality** states that:
"In any triangle, the sum of any two sides must always be greater than the third side."
Prove that $|\mathbf{a} + \mathbf{b}| \leqslant |\mathbf{a}| + |\mathbf{b}|$ using a geometrical argument.



Hint: Consider:
- **a** not parallel to **b** and use the triangle inequality
- **a** and **b** parallel
- any other cases.

14 Show that the following sets of points are collinear:

a A$(-1, 1)$, B$(4, 6)$, C$(1, 3)$
b P$(3, 3)$, Q$(6, -3)$, R$(1, 7)$
c A$(-2, 1, 4)$, B$(4, 3, 0)$, C$(19, 8, -10)$
d P$(2, 1, 1)$, Q$(5, -5, -2)$, R$(-1, 7, 4)$

15 Find a and b if:

a A$(2, -3, 4)$, B$(11, -9, 7)$, and C$(-13, a, b)$ are collinear
b K$(1, -1, 0)$, L$(4, -3, 7)$, and M$(a, 2, b)$ are collinear.

16 Let P(x, y, z) be a point on a sphere with radius r units and centre C(X, Y, Z).

a Write down the vector $\overrightarrow{\text{CP}}$.
b Use the distance formula for CP to write a Cartesian equation for the sphere.

INVESTIGATION 2 LINEAR COMBINATIONS

Any vector $\begin{pmatrix} r \\ s \end{pmatrix}$ in the plane can be written in the form $r\mathbf{i} + s\mathbf{j}$. We say that $\begin{pmatrix} r \\ s \end{pmatrix}$ is a **linear combination** of **i** and **j**.



Click on the icon to explore linear combinations further.

K THE SCALAR PRODUCT OF TWO VECTORS

For ordinary numbers a and b we can write the product of a and b as ab or $a \times b$. There is only one interpretation for this product, so we can use power notation $a^2 = a \times a$, $a^3 = a \times a \times a$, and so on as shorthand.

However, there are *two* different types of product involving two vectors. These are:

- The **scalar product** of 2 vectors, which results in a **scalar** answer and has the notation **a** • **b** (read "**a** dot **b**").
- The **vector product** of 2 vectors, which results in a **vector** answer and has the notation **a** × **b** (read "**a** cross **b**").

Consequently, for vector **a**, we do not use the terms $\mathbf{a}^2$ or $(\mathbf{a})^2$, as it is not clear which of the vector products it refers to.

For 2-dimensional vectors in a plane, we only need to talk about scalar product. The vector cross product will be important for 3-dimensional vectors in space.

SCALAR PRODUCT

The **scalar product** of two vectors is also known as the **dot product** or **inner product**.

► $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \bullet \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = a_1b_1 + a_2b_2$

► $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \bullet \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3$

Be careful not to confuse the **scalar product**, which is the product of two vectors to give a scalar answer, with **scalar multiplication**, which is the product of a scalar and a vector to give a parallel vector. They are quite different.

Example 28 Self Tutor

Find: **a** $\begin{pmatrix} 2 \\ 3 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ **b** $\begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$

a $\begin{pmatrix} 2 \\ 3 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 4 \end{pmatrix}$
$= 2 \times 1 + 3 \times 4$
$= 14$

b $\begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$
$= -1 \times 3 + 2 \times 1 + 4 \times 2$
$= 7$

ALGEBRAIC PROPERTIES OF THE SCALAR PRODUCT

1 $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$

2 $\mathbf{a} \bullet \mathbf{a} = |\mathbf{a}|^2$

3 $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

4 $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$

EXERCISE 12K

1 Find:

a $\begin{pmatrix} 2 \\ 5 \end{pmatrix} \bullet \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ **b** $\begin{pmatrix} 5 \\ -1 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ **c** $\begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix}$ **d** $\begin{pmatrix} -4 \\ -5 \\ 3 \end{pmatrix} \bullet \begin{pmatrix} 9 \\ -6 \\ -1 \end{pmatrix}$

2 For $\mathbf{p} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, find:

a $\mathbf{q} \bullet \mathbf{p}$ **b** $\mathbf{q} \bullet \mathbf{r}$ **c** $\mathbf{q} \bullet (\mathbf{p} + \mathbf{r})$ **d** $3\mathbf{r} \bullet \mathbf{q}$

e $2\mathbf{p} \bullet 2\mathbf{p}$ **f** $\mathbf{i} \bullet \mathbf{p}$ **g** $\mathbf{q} \bullet \mathbf{j}$ **h** $\mathbf{i} \bullet \mathbf{i}$

3 For $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$, find:

a $\mathbf{a} \bullet \mathbf{b}$ **b** $\mathbf{b} \bullet \mathbf{a}$ **c** $|\mathbf{a}|^2$

d $\mathbf{a} \bullet \mathbf{a}$ **e** $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c})$ **f** $\mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

4 Use the general vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ to prove that:

a $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$ **b** $\mathbf{a} \bullet \mathbf{a} = |\mathbf{a}|^2$

5 Find:

a $(\mathbf{i} + \mathbf{j} - \mathbf{k}) \bullet (2\mathbf{j} + \mathbf{k})$ **b** $\mathbf{i} \bullet \mathbf{i}$ **c** $(3\mathbf{i} - 2\mathbf{k}) \bullet (\mathbf{i} + \mathbf{j})$

d $(2\mathbf{i} + \mathbf{k}) \bullet (-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ **e** $\mathbf{i} \bullet \mathbf{j}$ **f** $(\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}) \bullet (3\mathbf{i} - 3\mathbf{j} - 5\mathbf{k})$

6 Let $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$, and $\mathbf{d} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$.

a Find:

i $\mathbf{a} \bullet \mathbf{c}$ **ii** $\mathbf{a} \bullet \mathbf{d}$ **iii** $\mathbf{b} \bullet \mathbf{c}$ **iv** $\mathbf{b} \bullet \mathbf{d}$

b Hence show that $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$.

7 Use $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ to prove that $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$.

Hence prove that $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$.

8 Use the properties of the scalar product to show that:

a $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + 2\mathbf{a} \bullet \mathbf{b} + |\mathbf{b}|^2$ **b** $|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2 = 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$

c $|\mathbf{a} + \mathbf{b}|^2 - |\mathbf{a} - \mathbf{b}|^2 = 4\mathbf{a} \bullet \mathbf{b}$

9 Explain why $\mathbf{a} \bullet \mathbf{b} \bullet \mathbf{c}$ is meaningless.

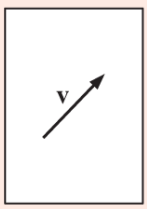
10 If $|\mathbf{a}| = 5$ and $|\mathbf{b}| = 3$, find $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{a} - \mathbf{b})$.

11 Show that if vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel, then $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}|$.

L THE ANGLE BETWEEN TWO VECTORS

DISCUSSION

1

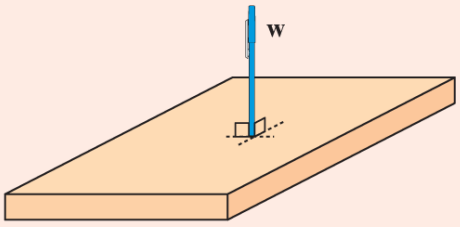


Elaine has drawn a vector $\mathbf{v}$ on a *plane* which is a sheet of paper. It is therefore a 2-dimensional vector.

a How many vectors can she draw which are *perpendicular* to $\mathbf{v}$?

b Are all of these vectors parallel?

2 Edward is thinking about vectors in *space*. These are 3-dimensional vectors. He is holding his pen vertically on his desk to represent a vector **w**.



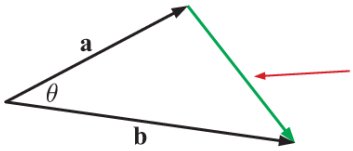
a How many vectors are there which are *perpendicular* to **w**?

b If Edward was to draw a vector (in pencil) on his desk, would it be perpendicular to **w**?

c Are all of the vectors which are perpendicular to **w**, parallel to one another?

THE ANGLE BETWEEN TWO VECTORS

Suppose θ is the angle between the vectors **a** and **b**.



This vector is $-\mathbf{a} + \mathbf{b}$ or $\mathbf{b} - \mathbf{a}$.
Its length is $|\mathbf{b} - \mathbf{a}|$.

Using the cosine rule, $|\mathbf{b} - \mathbf{a}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta$

Letting $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$,

$$\mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} - \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \\ b_3 - a_3 \end{pmatrix}$$

$\therefore\ (b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2 = (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2) - 2|\mathbf{a}||\mathbf{b}|\cos\theta$

$\therefore\ 2a_1b_1 + 2a_2b_2 + 2a_3b_3 = 2|\mathbf{a}||\mathbf{b}|\cos\theta$ {simplifying}

$\therefore\ a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$

$\therefore\ \mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$

The angle θ between two vectors **a** and **b** can be found using $\cos\theta = \dfrac{\mathbf{a} \bullet \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.

Example 29 Self Tutor

If $\mathbf{p} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$, find the angle between **p** and **q**.

$$\mathbf{p} \bullet \mathbf{q} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \bullet \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$
$$= 2(-1) + 3(0) + (-1)2$$
$$= -2 + 0 - 2$$
$$= -4$$

Now $\cos\theta = \dfrac{\mathbf{p} \bullet \mathbf{q}}{|\mathbf{p}||\mathbf{q}|}$

$$= \frac{-4}{\sqrt{4+9+1}\sqrt{1+0+4}}$$
$$= \frac{-4}{\sqrt{70}}$$

$\therefore\ \theta = \cos^{-1}\left(\frac{-4}{\sqrt{70}}\right) \approx 119°$

GEOMETRIC PROPERTIES OF THE SCALAR PRODUCT

Using the formula $\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$, we notice that:

► If $\mathbf{a}$ is perpendicular to $\mathbf{b}$ then $\theta = 90°$.

$$\therefore \quad \mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$
$$= |\mathbf{a}||\mathbf{b}|\cos 90°$$
$$= 0$$

► If $\mathbf{a}$ is parallel to $\mathbf{b}$ then $\theta = 0°$ or $180°$.

$$\therefore \quad \mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$
$$= |\mathbf{a}||\mathbf{b}|\cos 0° \quad \text{or} \quad |\mathbf{a}||\mathbf{b}|\cos 180°$$
$$= \pm|\mathbf{a}||\mathbf{b}|$$
$$\therefore \quad |\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}|$$

We are hence able to state the following **geometric properties of the scalar product**:

For non-zero vectors $\mathbf{a}$ and $\mathbf{b}$:

1 $\mathbf{a} \bullet \mathbf{b} = 0 \quad \Leftrightarrow \quad \mathbf{a}$ and $\mathbf{b}$ are **perpendicular**.

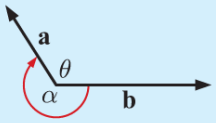
2 $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \quad \Leftrightarrow \quad \mathbf{a}$ and $\mathbf{b}$ are **parallel vectors**.

3 If θ is the angle between vectors $\mathbf{a}$ and $\mathbf{b}$ then $\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$.

If θ is acute, $\cos\theta > 0$ and so $\mathbf{a} \bullet \mathbf{b} > 0$

If θ is obtuse, $\cos\theta < 0$ and so $\mathbf{a} \bullet \mathbf{b} < 0$.

The angle between two vectors is always taken as the angle θ such that $0° \leqslant \theta \leqslant 180°$, rather than reflex angle α.



EXERCISE 12L

1 If $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, find:

a $\mathbf{a} \bullet \mathbf{b}$ **b** the angle between $\mathbf{a}$ and $\mathbf{b}$.

2 Find the angle between $\mathbf{r}$ and $\mathbf{s}$, given:

a $\mathbf{r} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **b** $\mathbf{r} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

c $\mathbf{r} = \mathbf{i} - \mathbf{j}$ and $\mathbf{s} = 2\mathbf{i} - \mathbf{j}$ **d** $\mathbf{r} = \mathbf{i}$ and $\mathbf{s} = \mathbf{i} + \mathbf{j}$

3 Show that $\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ are perpendicular.

4 Show that $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix}$ are mutually perpendicular.

Vectors are *mutually perpendicular* if each one is perpendicular to all the others.



5 Find the angle between:

a $\begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$

b $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$

c $2\mathbf{j} - \mathbf{k}$ and $\mathbf{i} + 2\mathbf{k}$

d $\mathbf{j}$ and $-\mathbf{i} + \mathbf{j} + 3\mathbf{k}$

6 Find t given that the following pairs of vectors are perpendicular:

a $\mathbf{p} = \begin{pmatrix} 3 \\ t \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

b $\mathbf{r} = \begin{pmatrix} t \\ t+2 \end{pmatrix}$, $\mathbf{s} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

c $\mathbf{a} = \begin{pmatrix} t \\ t+2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2-3t \\ t \end{pmatrix}$

d $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ t \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2t \\ -3 \\ -4 \end{pmatrix}$

e $\mathbf{p} = \begin{pmatrix} 3 \\ t \\ -2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 1-t \\ -3 \\ 4 \end{pmatrix}$

f $\mathbf{a} = 2\mathbf{i} + t\mathbf{j} + (t-2)\mathbf{k}$, $\mathbf{b} = t\mathbf{i} + 3\mathbf{j} + t\mathbf{k}$

7 Find r, s, and t such that $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2 \\ 2 \\ r \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} s \\ t \\ 1 \end{pmatrix}$ are mutually perpendicular.

Example 30 **Self Tutor**

Find a vector of length 3 units which is perpendicular to $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

$\begin{pmatrix} -1 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ 1 \end{pmatrix} = -4 + 4 = 0$

$\therefore$ vectors of the form $k\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ are perpendicular to $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

Now $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ has length $\sqrt{16+1} = \sqrt{17}$ units

$\therefore$ $k = \frac{1}{\sqrt{17}}$ for a unit vector.

$\therefore$ a vector of length 3 units is $\frac{3}{\sqrt{17}}\begin{pmatrix} 4 \\ 1 \end{pmatrix}$, which is $\begin{pmatrix} \frac{12}{\sqrt{17}} \\ \frac{3}{\sqrt{17}} \end{pmatrix}$.

8 Find a vector of length:

a 10 units which is perpendicular to $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$

b $3\sqrt{2}$ units which is perpendicular to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

c $\sqrt{20}$ units which is perpendicular to $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$.

9 Find *two* vectors of length 5 units which are perpendicular to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Example 31 **Self Tutor**

a and **b** have lengths 5 units and $\sqrt{7}$ units respectively, and the angle between them is $110°$. Find $\mathbf{a} \bullet \mathbf{b}$.

$$\begin{aligned} \mathbf{a} \bullet \mathbf{b} &= |\mathbf{a}||\mathbf{b}| \cos\theta \\ &= 5 \times \sqrt{7} \times \cos 110° \\ &\approx -4.525 \end{aligned}$$

10 **a** and **b** have lengths $\sqrt{10}$ units and 3 units respectively and the angle between them is $71°$. Find $\mathbf{a} \bullet \mathbf{b}$.

11 Find $\mathbf{p} \bullet \mathbf{q}$ for:

a $|\mathbf{p}| = 2$, $|\mathbf{q}| = 5$, and $\theta = 60°$ **b** $|\mathbf{p}| = 6$, $|\mathbf{q}| = 3$, and $\theta = \frac{2\pi}{3}$

12 Prove that two non-zero vectors **a** and **b** are perpendicular if and only if $\mathbf{a} \bullet \mathbf{b} = 0$.

Hint: You need to show that **a** and **b** being perpendicular implies $\mathbf{a} \bullet \mathbf{b} = 0$ *and* that $\mathbf{a} \bullet \mathbf{b} = 0$ implies **a** and **b** are perpendicular.

13 Prove that if **a** and **b** are non-zero vectors then $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \Leftrightarrow$ **a** and **b** are parallel.

14 Suppose $\mathbf{a} \bullet \mathbf{b} = -12$ and **b** is a unit vector.

a Explain why **a** and **b** are not perpendicular.

b Given that **a** and **b** are parallel, find $|\mathbf{a}|$.

15 Suppose $|\mathbf{c}| = |\mathbf{d}| = \sqrt{5}$. What can be deduced about **c** and **d** if:

a $\mathbf{c} \bullet \mathbf{d} = 5$ **b** $\mathbf{c} \bullet \mathbf{d} = -5$?

16 Find three vectors **a**, **b**, **c** in space such that $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{a} \bullet \mathbf{b} = \mathbf{a} \bullet \mathbf{c}$ but $\mathbf{b} \neq \mathbf{c}$.

17 Find any two vectors which are not parallel, but which are *both* perpendicular to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

18 Consider triangle ABC in which A is $(5, 1, 2)$, B is $(6, -1, 0)$, and C is $(3, 2, 0)$. Using scalar product only, show that the triangle is right angled.

19 A$(2, 4, 2)$, B$(-1, 2, 3)$, C$(-3, 3, 6)$, and D$(0, 5, 5)$ are vertices of a quadrilateral.

a Prove that ABCD is a parallelogram.

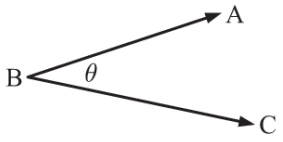
b Find $|\overrightarrow{AB}|$ and $|\overrightarrow{BC}|$. What can be said about ABCD?

c Find $\overrightarrow{AC} \bullet \overrightarrow{BD}$ and discuss your answer.

20 Find the angle ABC of triangle ABC for A$(3, 0, 1)$, B$(-3, 1, 2)$, and C$(-2, 1, -1)$.

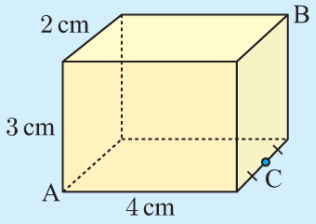
Hint: To find the angle at B, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$.

What angle is found if $\overrightarrow{BA}$ and $\overrightarrow{CB}$ are used?

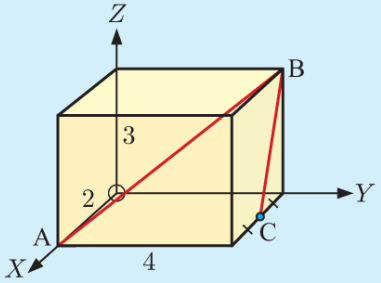


Example 32 Self Tutor

Use vector methods to determine the measure of $A\widehat{B}C$.



Placing the coordinate axes as illustrated, A is $(2, 0, 0)$, B is $(0, 4, 3)$, and C is $(1, 4, 0)$.



$$\therefore \ \overrightarrow{BA} = \begin{pmatrix} 2 \\ -4 \\ -3 \end{pmatrix} \text{ and } \overrightarrow{BC} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$$

Now $$\cos(A\widehat{B}C) = \frac{\overrightarrow{BA} \bullet \overrightarrow{BC}}{|\overrightarrow{BA}| \ |\overrightarrow{BC}|}$$

$$= \frac{2(1) + (-4)(0) + (-3)(-3)}{\sqrt{4+16+9}\sqrt{1+0+9}}$$

$$= \frac{11}{\sqrt{290}}$$

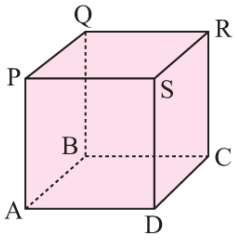
$$\therefore \ A\widehat{B}C = \cos^{-1}\left(\frac{11}{\sqrt{290}}\right) \approx 49.8°$$

The vectors used must both be away from B (or towards B). If this is not done you will be finding the exterior angle at B.



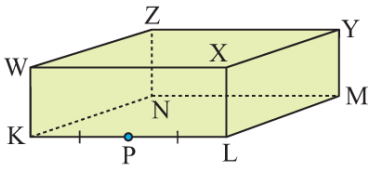
21 The cube alongside has edges of length 2 cm. Find, using vector methods, the measure of:

a $A\widehat{B}S$ **b** $R\widehat{B}P$ **c** $P\widehat{B}S$



22 [KL], [LM], and [LX] are 8, 5, and 3 units long respectively. P is the midpoint of [KL]. Find, using vector methods, the measure of:

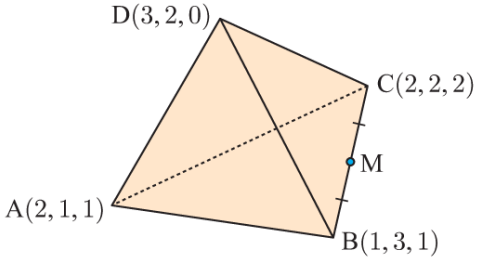
a $Y\widehat{N}X$ **b** $Y\widehat{N}P$



23 Consider the tetrahedron ABCD.

a Find the coordinates of M.

b Find the measure of $D\widehat{M}A$.

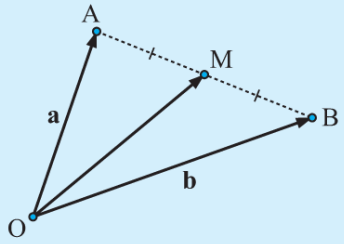


M PROOF USING VECTOR GEOMETRY

We can use vectors to prove important geometrical facts. The use of vectors in such proofs provides us with an alternative to using deductive or coordinate arguments, and can often simplify the proofs.

Example 33 **Self Tutor**

If M is the midpoint of [AB], prove that $\overrightarrow{OM} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$.



$$\begin{aligned}\overrightarrow{OM} &= \overrightarrow{OA} + \overrightarrow{AM} \\ &= \overrightarrow{OA} + \tfrac{1}{2}\overrightarrow{AB} \\ &= \mathbf{a} + \tfrac{1}{2}(\mathbf{b} - \mathbf{a}) \\ &= \tfrac{1}{2}\mathbf{a} + \tfrac{1}{2}\mathbf{b}\end{aligned}$$

USEFUL TOOLS IN VECTOR PROOF

- If $\mathbf{a} = k\mathbf{b}$ where k is a scalar then $\mathbf{a}$ and $\mathbf{b}$ are parallel, and $|\mathbf{a}| = |k||\mathbf{b}|$.
- If M is the midpoint of [AB] then $\overrightarrow{OM} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$.
- To prove $\mathbf{a}$ is perpendicular to $\mathbf{b}$, show that $\mathbf{a} \bullet \mathbf{b} = 0$.
- Properties of scalar product:
  1. $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$ for any two vectors $\mathbf{a}$ and $\mathbf{b}$.
  2. $\mathbf{a} \bullet \mathbf{a} = |\mathbf{a}|^2$ for any vector $\mathbf{a}$.
  3. $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$ for any vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
  4. $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$.

Example 34 **Self Tutor**

Prove that the line joining the apex to the midpoint of the base of an isosceles triangle is perpendicular to the base.

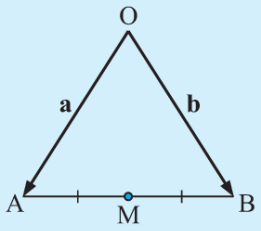
Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

$\therefore \overrightarrow{AB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$

and $\overrightarrow{OM} = \mathbf{a} + \frac{1}{2}\overrightarrow{AB}$

$= \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a})$

$= \frac{1}{2}(\mathbf{a} + \mathbf{b})$



$\therefore \overrightarrow{OM} \bullet \overrightarrow{AB} = \frac{1}{2}(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{b} - \mathbf{a})$

$= \frac{1}{2}(\mathbf{a} \bullet \mathbf{b} - \mathbf{a} \bullet \mathbf{a} + \mathbf{b} \bullet \mathbf{b} - \mathbf{b} \bullet \mathbf{a})$

$= \frac{1}{2}(\mathbf{a} \bullet \mathbf{b} - \mathbf{b} \bullet \mathbf{a})$

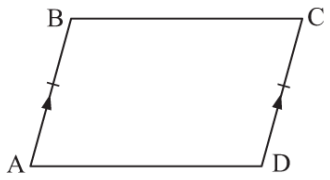
$= 0$

$\therefore \overrightarrow{OM} \perp \overrightarrow{AB}$

EXERCISE 12M

1 Prove that: *If a pair of opposite sides of a quadrilateral are parallel and equal in length, then the quadrilateral is a parallelogram.*

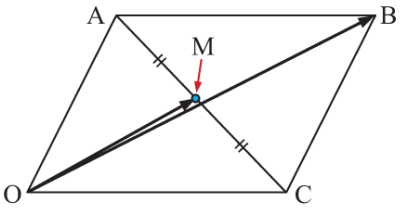
Hint:



Let $\overrightarrow{AB} = \mathbf{r}$ and $\overrightarrow{AD} = \mathbf{s}$.
Find $\overrightarrow{DC}$ and $\overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DC}$ in terms of $\mathbf{r}$ and $\mathbf{s}$.

2 Prove that: *The diagonals of a parallelogram bisect each other.*

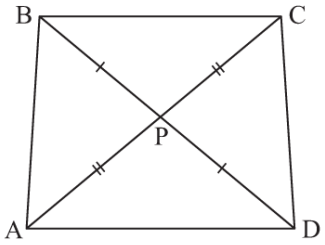
Hint:



M is the midpoint of [AC].
Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.
Find $\overrightarrow{OB}$ and $\overrightarrow{OM}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

3 Prove that: *The line joining the midpoints of two sides of a triangle is parallel to the third side and half its length.*

4 ABCD is a quadrilateral in which BP = PD and AP = PC.
Let $\overrightarrow{AP} = \mathbf{r}$ and $\overrightarrow{PB} = \mathbf{s}$.



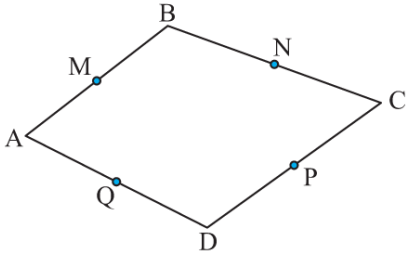
a Find, in terms of $\mathbf{r}$ and $\mathbf{s}$:

i $\overrightarrow{PC}$ and $\overrightarrow{DP}$ **ii** $\overrightarrow{AB}$ and $\overrightarrow{DC}$.

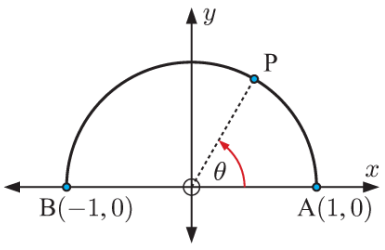
b Discuss the significance of your answers to **a ii**.

c Copy and complete: *If the diagonals of a quadrilateral bisect each other .....*

5 Consider a quadrilateral ABCD where the midpoints of the sides are M, N, P, and Q as shown.
Prove that MNPQ is a parallelogram.



6 In the given figure:



a State the coordinates of P.

b Find $\overrightarrow{BP}$ and $\overrightarrow{AP}$.

c Find $\overrightarrow{AP} \bullet \overrightarrow{BP}$ using **b**.

d What property of a semi-circle has been deduced in **c**?

7 $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of two distinct points A and B, neither of which is the origin. Show that if $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$ then $\mathbf{a}$ is perpendicular to $\mathbf{b}$, using:

a a vector algebraic method **b** a geometric argument.

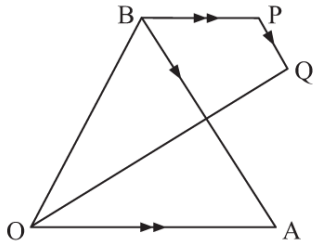
8 Prove that the diagonals of a rhombus bisect the angles of the rhombus using:

a a geometric argument **b** vector methods.

9 Prove that: *The diagonals of a parallelogram meet at right angles if and only if it is a rhombus.*
Hint: For an "if and only if" proof, you need to prove it in both directions.

10 Triangle OAB is equilateral. [BP] is parallel to [OA] and a half of its length. [PQ] is parallel to [BA] and a quarter of its length.

Let $\overrightarrow{OA} = \mathbf{r}$ and $\overrightarrow{OB} = \mathbf{s}$.



a Find vector expressions for $\overrightarrow{AB}$ and $\overrightarrow{OQ}$ in terms of $\mathbf{r}$ and $\mathbf{s}$.

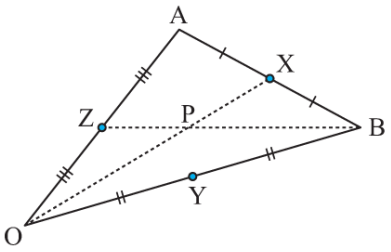
b Prove that [AB] is perpendicular to [OQ].

11 Suppose $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors. Prove that if $k_1\mathbf{a} + k_2\mathbf{b} = l_1\mathbf{a} + l_2\mathbf{b}$, then $k_1 = l_1$ and $k_2 = l_2$.

12 Prove that: *The sum of the squares of the lengths of the diagonals of a parallelogram is equal to the sum of the squares of the lengths of its sides.*

13 Consider a triangle OAB where the midpoints of the sides are X, Y, and Z as shown. [OX], [AY], and [BZ] are called the *medians* of the triangle.

Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.



a Write $\overrightarrow{OX}$ and $\overrightarrow{BZ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

b Suppose [OX] and [BZ] intersect at P. Write $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

c Write $\overrightarrow{YP}$ and $\overrightarrow{PA}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

d Show that the medians of a triangle meet at a common point.

N THE VECTOR PRODUCT OF TWO VECTORS

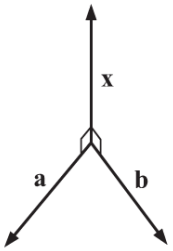
We have seen how the scalar product of two vectors results in a **scalar**.

The second form of product between two vectors is the **vector product** or **vector cross product**, and this results in a **vector**.

The vector cross product is only defined for 3-dimensional vectors.

The vector product arises when we attempt to find a vector which is *perpendicular to two other known vectors*.

Suppose $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is perpendicular to both $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.



$$\therefore \begin{cases} a_1x + a_2y + a_3z = 0 \\ b_1x + b_2y + b_3z = 0 \end{cases} \quad \{\text{as dot products are zero}\}$$

$$\therefore \begin{cases} a_1x + a_2y = -a_3z & \ldots. (1) \\ b_1x + b_2y = -b_3z & \ldots. (2) \end{cases}$$

We will now try to solve these two equations simultaneously to obtain expressions for x and y in terms of z.

To eliminate x, we multiply (1) by $-b_1$ and (2) by a_1.

$$\begin{array}{rll} -a_1b_1x - a_2b_1y = a_3b_1z & & \{-b_1 \times (1)\} \\ a_1b_1x + a_1b_2y = -a_1b_3z & & \{a_1 \times (2)\} \\ \hline (a_1b_2 - a_2b_1)y = (a_3b_1 - a_1b_3)z & & \end{array}$$

Adding these gives

$\therefore \quad y = (a_3b_1 - a_1b_3)t$ and $z = (a_1b_2 - a_2b_1)t$ for any non-zero t.

Substituting into (1), $\quad a_1x + a_2(a_3b_1 - a_1b_3)t = -a_3(a_1b_2 - a_2b_1)t$

$$\therefore \quad a_1x = (-a_1a_3b_2 + \cancel{a_2a_3b_1} - \cancel{a_2a_3b_1} + a_1a_2b_3)t$$

$$\therefore \quad a_1x = a_1(a_2b_3 - a_3b_2)t$$

$$\therefore \quad x = (a_2b_3 - a_3b_2)t$$

Letting $t = 1$, we obtain $\mathbf{x} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$, and we call this vector the **vector cross product** of **a** and **b**.

The **vector cross product** of vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$.

A formula like this will be given in your exam formula booklet. However, the formula may be confusing to use if, for example, you were finding $\mathbf{b} \times \mathbf{a}$. For this reason we recommend an alternative method for calculating vector cross product. We will discover it in the following **Investigation**.

INVESTIGATION 3 THE VECTOR CROSS PRODUCT FORMULA

In matrix algebra, we can define a **determinant** for any square matrix. In particular:

- The determinant of the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

  It is the product of the terms on the main (red) diagonal minus the product of the other terms.

  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

- The determinant of the 3×3 matrix $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ is

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Notice that:

- The sign of the middle term is negative.
- The *size* of the ith term is a_{1i} times the 2×2 determinant seen by covering the top row and the ith column.

VIDEO

What to do:

Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ be vectors.

Show that $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$

is equal to the vector cross product $\mathbf{a} \times \mathbf{b}$.

From the **Investigation**, you should have found that:

The vector cross product of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$



Example 35 **Self Tutor**

For $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}$, find:

a $\mathbf{b} \times \mathbf{c}$ **b** $\mathbf{a} \bullet (\mathbf{b} \times \mathbf{c})$

a $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 2 & 0 & 4 \end{vmatrix}$

$= \begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} \mathbf{k}$

$= \begin{pmatrix} 8 \\ 2 \\ -4 \end{pmatrix}$

or $\mathbf{b} \times \mathbf{c} = \begin{pmatrix} b_2c_3 - b_3c_2 \\ b_3c_1 - b_1c_3 \\ b_1c_2 - b_2c_1 \end{pmatrix}$

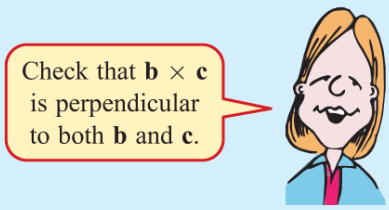
$= \begin{pmatrix} (2)(4) - (3)(0) \\ (3)(2) - (1)(4) \\ (1)(0) - (2)(2) \end{pmatrix}$

$= \begin{pmatrix} 8 \\ 2 \\ -4 \end{pmatrix}$

b $\mathbf{a} \bullet (\mathbf{b} \times \mathbf{c}) = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \bullet \begin{pmatrix} 8 \\ 2 \\ -4 \end{pmatrix}$

$= 16 + 2 + 4$

$= 22$



ALGEBRAIC PROPERTIES OF THE VECTOR CROSS PRODUCT

The vector cross product has the following algebraic properties for non-zero 3-dimensional vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$:

1. $\mathbf{a} \times \mathbf{b}$ is a vector which is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
2. $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for all $\mathbf{a}$.
3. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ for all $\mathbf{a}$ and $\mathbf{b}$.
   This means that $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$ have the same length but opposite direction.
4. $\mathbf{a} \bullet (\mathbf{b} \times \mathbf{c})$ is called the **scalar triple product**.
5. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
6. $(\mathbf{a} + \mathbf{b}) \times (\mathbf{c} + \mathbf{d}) = (\mathbf{a} \times \mathbf{c}) + (\mathbf{a} \times \mathbf{d}) + (\mathbf{b} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{d})$.

You will prove or verify these results in the next Exercise.

EXERCISE 12N.1

1 Calculate:

a $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$ **b** $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$ **c** $\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -3 \\ 1 \\ -5 \end{pmatrix}$

d $(\mathbf{i} + \mathbf{j} - 2\mathbf{k}) \times (\mathbf{i} - \mathbf{k})$ **e** $(2\mathbf{i} - \mathbf{k}) \times (\mathbf{j} + 3\mathbf{k})$ **f** $(\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \times (-2\mathbf{i} + 3\mathbf{j} - \mathbf{k})$

2 Suppose $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ -1 \end{pmatrix}$.

a Find $\mathbf{a} \times \mathbf{b}$. **b** Hence determine $\mathbf{a} \bullet (\mathbf{a} \times \mathbf{b})$ and $\mathbf{b} \bullet (\mathbf{a} \times \mathbf{b})$.

c Explain your results.

3 $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the base unit vectors parallel to the coordinate axes.

a Find $\mathbf{i} \times \mathbf{i}$, $\mathbf{j} \times \mathbf{j}$, and $\mathbf{k} \times \mathbf{k}$. Comment on your results.

b Find $\mathbf{i} \times \mathbf{j}$ and $\mathbf{j} \times \mathbf{i}$. Comment on your results.

c Predict what the following vector cross products will be, then check your prediction:

i $\mathbf{j} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{j}$ **ii** $\mathbf{i} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{i}$.

4 Use $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ to prove that:

a $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for all 3-dimensional vectors $\mathbf{a}$

b $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ for all 3-dimensional vectors $\mathbf{a}$ and $\mathbf{b}$.

5 Suppose $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$. Find:

a $\mathbf{b} \times \mathbf{c}$ **b** $\mathbf{a} \bullet (\mathbf{b} \times \mathbf{c})$

6 Suppose $\mathbf{a} = \mathbf{i} + 2\mathbf{k}$, $\mathbf{b} = -\mathbf{j} + \mathbf{k}$, and $\mathbf{c} = 2\mathbf{i} - \mathbf{k}$. Find:

a $\mathbf{a} \times \mathbf{b}$ **b** $\mathbf{a} \times \mathbf{c}$

c $(\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$ **d** $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$

7 **a** Use $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ to prove that $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

b *Hence* prove that $(\mathbf{a} + \mathbf{b}) \times (\mathbf{c} + \mathbf{d}) = (\mathbf{a} \times \mathbf{c}) + (\mathbf{a} \times \mathbf{d}) + (\mathbf{b} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{d})$.

Hint: Since $\mathbf{x} \times \mathbf{y} = -(\mathbf{y} \times \mathbf{x})$, the order of the vectors must be maintained.

8 Use the properties of vector cross product to simplify:

a $\mathbf{a} \times (\mathbf{a} + \mathbf{b})$ **b** $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{b})$

c $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b})$ **d** $2\mathbf{a} \bullet (\mathbf{a} \times \mathbf{b})$

9 **a** Use general vectors to show that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \bullet \mathbf{c})\mathbf{b} - (\mathbf{a} \bullet \mathbf{b})\mathbf{c}$.

b Hence show that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$.

Example 36 **Self Tutor**

Find *all* vectors perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ 1 & 0 & -3 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -1 \\ 0 & -3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & -1 \\ 1 & -3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} \mathbf{k}$$

$$= -6\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$$

$$= -2(3\mathbf{i} - \mathbf{j} + \mathbf{k})$$

or

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

$$= \begin{pmatrix} (2)(-3) - (-1)(0) \\ (-1)(1) - (1)(-3) \\ (1)(0) - (2)(1) \end{pmatrix}$$

$$= \begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix}$$

$$= -2 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

The vectors have the form $k \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$ where k is any non-zero real number.

10 Find *all* vectors perpendicular to both:

a $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ **b** $\begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix}$

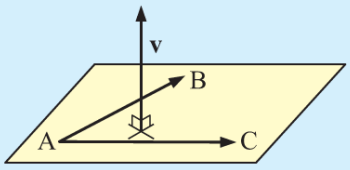
c $\mathbf{i} + \mathbf{j}$ and $\mathbf{i} - \mathbf{j} - \mathbf{k}$ **d** $\mathbf{i} - \mathbf{j} - \mathbf{k}$ and $2\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$.

11 Find all vectors perpendicular to both $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.

Hence find *two* vectors of length 5 units which are perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

Example 37 **Self Tutor**

Find a vector which is perpendicular to the plane passing through the points A(1, −1, 2), B(3, 1, 0), and C(−1, 2, −3).



$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -2 \\ 3 \\ -5 \end{pmatrix}$

The vector $\mathbf{v}$ must be perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

$$\therefore \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -2 \\ -2 & 3 & -5 \end{vmatrix} = \begin{vmatrix} 2 & -2 \\ 3 & -5 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & -2 \\ -2 & -5 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 2 \\ -2 & 3 \end{vmatrix}\mathbf{k}$$

$$= -4\mathbf{i} + 14\mathbf{j} + 10\mathbf{k}$$

$$= -2(2\mathbf{i} - 7\mathbf{j} - 5\mathbf{k})$$

Any non-zero multiple of $\begin{pmatrix} 2 \\ -7 \\ -5 \end{pmatrix}$ will be perpendicular to the plane.

12 Find a vector which is perpendicular to the plane passing through the points:

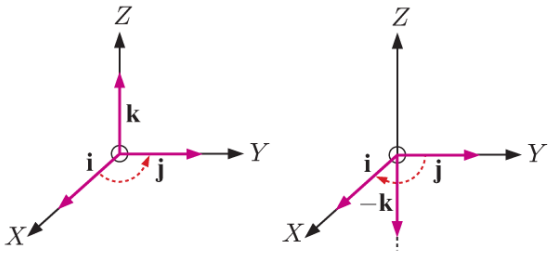
a A(1, 3, 2), B(0, 2, −5), C(3, 1, −4) **b** P(2, 0, −1), Q(0, 1, 3), R(1, −1, 1)

THE DIRECTION OF $\mathbf{a} \times \mathbf{b}$

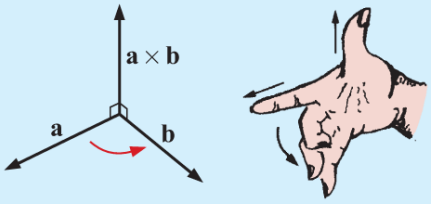
We have already observed that $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$, so $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$ are in opposite directions.

However, what is the direction of each?

In the last Exercise, we saw that $\mathbf{i} \times \mathbf{j} = \mathbf{k}$ and $\mathbf{j} \times \mathbf{i} = -\mathbf{k}$.



In general, the **direction** of $\mathbf{a} \times \mathbf{b}$ is determined by the **right hand rule**:



If the fingers on your right hand turn from $\mathbf{a}$ to $\mathbf{b}$, then your thumb points in the direction of $\mathbf{a} \times \mathbf{b}$.

THE LENGTH OF $\mathbf{a} \times \mathbf{b}$

Using $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$, $|\mathbf{a} \times \mathbf{b}| = \sqrt{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2}$.

However, another very useful form of the length of $\mathbf{a} \times \mathbf{b}$ exists:

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta \quad \text{where } \theta \text{ is the angle between } \mathbf{a} \text{ and } \mathbf{b}.$$

Proof:

$$\begin{aligned}
|\mathbf{a}|^2 |\mathbf{b}|^2 \sin^2\theta &= |\mathbf{a}|^2 |\mathbf{b}|^2 (1 - \cos^2\theta) \\
&= |\mathbf{a}|^2 |\mathbf{b}|^2 - |\mathbf{a}|^2 |\mathbf{b}|^2 \cos^2\theta \\
&= |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \bullet \mathbf{b})^2 \\
&= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\
&\qquad \text{which on expanding and then factorising gives} \\
&= (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \\
&= |\mathbf{a} \times \mathbf{b}|^2 \\
\therefore \quad |\mathbf{a} \times \mathbf{b}| &= |\mathbf{a}||\mathbf{b}| \sin\theta \qquad \{\text{as} \quad \sin\theta > 0\}
\end{aligned}$$

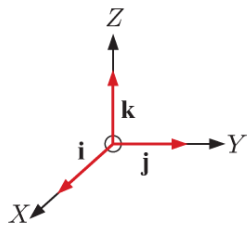
The immediate consequences of this result are:

- If $\mathbf{u}$ is a **unit vector** in the direction of $\mathbf{a} \times \mathbf{b}$ then $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin\theta\, \mathbf{u}$. In some texts this is the **geometric definition** of $\mathbf{a} \times \mathbf{b}$.
- If $\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors, then $\mathbf{a} \times \mathbf{b} = \mathbf{0} \Leftrightarrow \mathbf{a}$ is parallel to $\mathbf{b}$.

EXERCISE 12N.2

1 **a** Find $\mathbf{i} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{i}$ using the original definition of $\mathbf{a} \times \mathbf{b}$. Check that the **right hand rule** correctly gives the direction of $\mathbf{i} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{i}$.

b Verify that $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin\theta\, \mathbf{u}$, where $\mathbf{u}$ is the unit vector in the direction of $\mathbf{a} \times \mathbf{b}$, is true for $\mathbf{i} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{i}$.



2 Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - \mathbf{k}$.

a Find $\mathbf{a} \bullet \mathbf{b}$ and $\mathbf{a} \times \mathbf{b}$.

b Find $\cos\theta$ using $\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$.

c Find $\sin\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

d Find $\sin\theta$ using $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$.

3 Prove the property:
"If $\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors then $\mathbf{a} \times \mathbf{b} = \mathbf{0} \Leftrightarrow \mathbf{a}$ is parallel to $\mathbf{b}$."

4 Suppose $\mathbf{p} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$.

a Show that $\mathbf{p}$ and $\mathbf{q}$ are perpendicular.

b Find $\mathbf{p} \times \mathbf{q}$.

c Verify that $|\mathbf{p} \times \mathbf{q}| = |\mathbf{p}||\mathbf{q}|$.

5 A, B, and C are 3 distinct points with non-zero position vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ respectively.

a If $\mathbf{a} \times \mathbf{c} = \mathbf{b} \times \mathbf{c}$, what can be deduced about $\overrightarrow{OC}$ and $\overrightarrow{AB}$?

b If $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, what relationship exists between $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{c}$?

c If $\mathbf{c} \neq \mathbf{0}$ and $\mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$, prove that $\mathbf{a} + \mathbf{b} = k\mathbf{c}$ for some scalar k.